Foundation-Model Prior Fusion for Low-Count PET Reconstruction with Multi-View VLM Prompting

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Abstract

Low-count positron emission tomography (PET) reconstruction is a low-quality-data problem in which fewer detections increase Poisson variability and anatomical guidance may be unavailable. We propose a PET-only method in which a vision-language Agent jointly processes grayscale and pseudocolour views of a preliminary maximum-likelihood expectation-maximization reconstruction and returns validated boxes and points for a PET-adapted MedSAM2 decoder. The continuous decoder output defines sparse local affinities for kernel expectation maximization (KEM); it is never blended into the reconstructed image, and the measured sinogram retains explicit Poisson data fidelity. We compare subject-disjoint BrainWeb and HECKTOR development-validation cohorts at 20K, 50K, 100K and 200K expected counts using subject-balanced aggregation, retaining higher-count observations as appendix references. At 20K, the routed pipeline achieves NRMSE/PSNR/SSIM of 0.3503/27.19 dB/0.8070 on revised BrainWeb and 0.2467/36.34 dB/0.9177 on HECKTOR. It leads the tested HECKTOR comparators across all four counts; on BrainWeb it exceeds the independent CNN-BPNet adaptation at 100K/200K, but not aligned MR guidance, and PET ADMM-Net retains higher SSIM. Equal-budget decoder continuation separates structural supervision from extra training. The Agent effect depends on the decoder: revised BrainWeb SSIM is lower than with matched deterministic prompts at every routed count, whereas HECKTOR 20K improves by 0.00264 SSIM with exploratory multiplicity-corrected patient-level support. Historical component evidence is reported separately rather than transferred to the revised decoder. These results support PET-derived prior fusion and dataset-dependent prompting, not universal Agent gains or externally validated generalization.

Keywords: low-count PET, PET reconstruction, information fusion, kernel expectation maximization, medical foundation model, vision-language agent

1. Introduction

Positron emission tomography (PET) measures radiotracer activity through detected coincidence events and supports diagnosis, staging, and treatment monitoring [24]. Reducing injected activity or acquisition time decreases the number of detected events. The resulting low-count measurements have stronger Poisson variability, and unconstrained maximum-likelihood reconstruction amplifies that variability as iterations proceed [3]. This is a quality-varying reconstruction problem: the observation statistics change with the available counts, yet clinically relevant uptake boundaries and small lesions must remain recoverable.

Anatomically guided PET reconstruction combines tracer measurements with co-registered magnetic resonance (MR) or computed tomography (CT) information. In kernel expectation maximization (KEM), similarities in a prior image define the reconstruction basis [4, 9], while parallel-level-set penalties offer a distinct anatomically guided formulation [6]. These approaches can improve noise control, but they require an additional image and reliable correspondence between modalities. When such anatomy is unavailable, treating it as an assumed second view makes the reconstruction system incomplete at deployment. We therefore ask whether a PET-derived foundation-model prior can supply the kernel representation while the original low-quality sinogram continues to determine data fidelity.

SAM-KEM addresses this question by coupling a learned prior branch to likelihood-based reconstruction. A preliminary maximum-likelihood expectation-maximization (MLEM) image supplies the visual context for prompting an adapted MedSAM2 decoder, and the resulting continuous PET prior defines local radial-basis-function (RBF) affinities for KEM. It is not an independent anatomical image: it is derived from the low-count PET input and trained against reference PET. The method therefore fuses two dependent representations with different roles—the sinogram supplies measured counts, whereas the learned image supplies a reconstruction basis. The KEM estimate is the final reconstruction. Motivated by the weak prompt-location effect in a historical matched analysis, we further test a dataset-aware VLM Agent that views only the noisy preliminary PET in grayscale and jet pseudocolour and returns strict per-object boxes and points for MedSAM2. The two renderings are presentation views of the same data: they can change visual salience for the Agent but cannot add independent measured information.

This design creates an auditable fusion chain for low-quality data: display views determine prompt objects, prompt objects condition the foundation prior, the prior determines local affini-

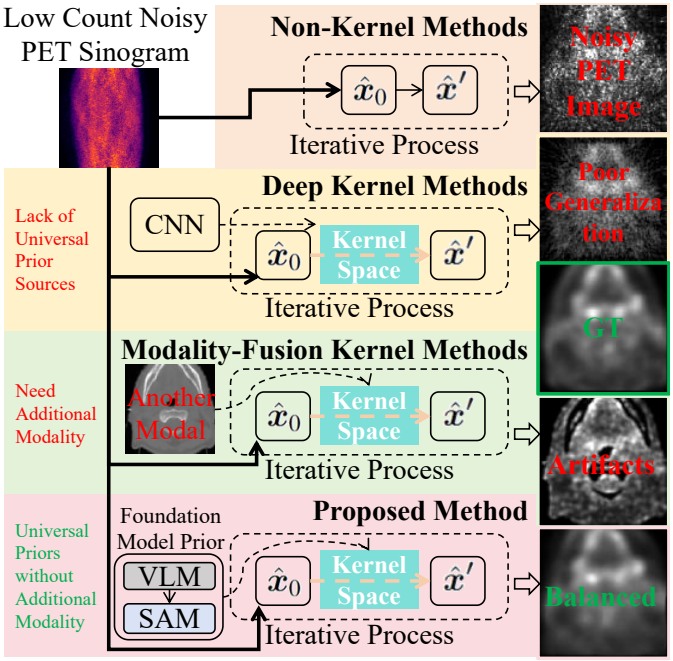


Figure 1: Conceptual comparison of reconstruction families for low-quality, count-limited PET data. Non-kernel methods directly iterate on the measured data; deep-kernel methods learn a PET-derived kernel; modality-guided kernel methods require an additional registered image; and the proposed approach constructs a foundation-model prior from PET itself before likelihood-based kernel reconstruction. The diagram is conceptual, and the quantitative comparisons use the common protocol described in Section 4.2.

ties, and the sparse kernel changes the parameterization optimized against the measured counts.

Figure 1 positions this PET-only prior-guided formulation relative to the principal low-count PET reconstruction families.

The contributions are:

1. a foundation-model prior fusion formulation in which a continuous PET-adapted MedSAM2 prior defines a sparse local KEM basis while the low-count sinogram remains in the Poisson likelihood and no prior image is blended into the output;
2. a dataset-aware, multi-view VLM prompt interface that jointly processes paired grayscale and pseudocolour presentations of preliminary PET and returns validated object-level prompts under a deterministic fallback policy; and
3. a subject-disjoint, multi-count evaluation that separates reconstruction quality, dose response, prior and kernel components, prompt-policy effects, and Agent reliability under frozen hyperparameters and intention-to-treat analysis.

2. Related Work

2.1. Classical PET Image Reconstruction

PET image reconstruction is fundamentally an inverse problem: estimating the tracer distribution $\boldsymbol{x}$ from noisy sinogram measurements $\boldsymbol{y}$. The statistical nature of PET data—governed by Poisson counting statistics—has motivated likelihood-based approaches since the seminal work of Shepp and Vardi [1], who derived the Expectation-Maximization (EM) algorithm for maximum likelihood estimation (MLEM). While MLEM converges to the maximum likelihood solution, it suffers from noise amplification at higher iterations, necessitating early stopping as an implicit regularization strategy.

The Ordered Subsets EM (OSEM) algorithm [2] accelerates MLEM by updating with subsets of the projection data. Both MLEM and OSEM amplify noise when iterated without regularization.

Penalized likelihood (PL) methods address this by adding explicit regularization terms to the objective function [3]:

$$\hat{\boldsymbol{x}} = \arg\max_{\boldsymbol{x}\geq 0} \left[L(\boldsymbol{y} \mid \boldsymbol{x}) - \beta R(\boldsymbol{x}) \right], \tag{1}$$

where $R(\boldsymbol{x})$ is a penalty and β controls its strength. Common choices include quadratic roughness, edge-preserving, and total-variation penalties. Anatomically guided variants derive neighbourhoods or gradient weights from co-registered MRI or CT. The Bowsher prior [5], for example, retains neighbours that are most similar in the anatomical image; parallel level sets [6] couple PET and anatomical gradient directions. These methods require careful parameter selection when the PET data are weak.

2.2. Kernel Methods in PET

Kernel-based PET reconstruction [4] parameterizes the image as $\boldsymbol{x} = \boldsymbol{K}\boldsymbol{\alpha}$, where $\boldsymbol{K}$ is constructed from image features and $\boldsymbol{\alpha}$ is estimated by EM. RBF similarities and neighbourhood sparsification make nearby pixels with similar features share information. This changes the reconstruction basis but does not, by itself, imply that $\boldsymbol{K}$ is low rank.

Anatomically guided kernel EM builds the kernel from co-registered MRI features, leveraging complementary PET–MRI structure [4, 9]. Regularized KEM adds a composite kernel-coefficient regularizer [7], whereas spatiotemporal kernels exploit correlations across dynamic PET frames [8].

Neural KEM [10] adds a deep coefficient prior to the kernel representation. Our method instead keeps coefficient-space EM and changes the image used to construct the kernel affinities.

2.3. Deep Learning for PET Reconstruction

Deep learning approaches to PET reconstruction fall into three categories: (i) post-processing denoising, (ii) end-to-end sinogram-to-image mapping, and (iii) unrolled iterative reconstruction networks.

Post-processing methods apply CNNs to OSEM or MLEM images. Their data consistency depends on whether the measurement model is included during training or inference.

Direct methods map sinograms to images. CNN-BPNet [11] combines back-projection layers with convolutional processing, and FBP-Net [12] combines filtered back-projection with learned components.

Unrolled approaches alternate measurement-model updates with learned regularization. ADMM-Net [13] follows this pattern using a fixed number of ADMM-like stages.

Score-based PET methods [14, 15] combine learned image distributions with iterative sampling or likelihood updates. Their computational cost and data-consistency mechanisms vary across implementations.

Wavelet-informed diffusion models have recently been proposed to reduce the cost of full-resolution diffusion. WiD-PET [16] applies a wavelet transform, a high-frequency enhancer, and spatial-consistency modules to reconstruct standard-dose PET from low-dose inputs; its authors report an inference time approximately 10% of their DDPM baseline. We discuss it as recent related work but do not transfer its performance numbers or present it as a matched baseline under our protocol.

2.4. Foundation Models in Medical Imaging

The Segment Anything Model (SAM) [17] provides prompt-conditioned masks but was trained on natural images. Medical benchmarks report variable performance without domain adaptation and show sensitivity to prompt choice [18, 19].

SAM-Med2D [20] and MedSAM2 [21] adapt promptable segmentation architectures to medical images. Their use as PET priors is not automatic, however, because reconstruction requires a continuous intensity representation rather than a binary semantic mask.

Vision-language models (VLMs) such as Qwen3-VL [22] and LLaVA-Med [23] accept images and text instructions. We use a hosted Qwen3.7-Max Agent only to generate MedSAM2 prompt objects from PET displays; the VLM neither reconstructs PET nor supplies an independently acquired information view. The earlier Qwen3-VL-Plus single-box path is retained as a historical comparator that established the adapted-prior and kernel components before the Agent policy was frozen.

2.5. Multi-View Foundation Priors for Low-Quality Reconstruction

Multi-view methods combine representations that differ in source, scale, transformation, or noise process [27, 28, 29]. Fusion may occur at the input, representation, decision, or output level [30]. Recent low-quality-data work distinguishes noisy, incomplete, imbalanced, and quality-varying fusion settings and emphasizes the reliability of foundation-model-assisted fusion under degradation [31]. Anatomically guided KEM is a feature-level example: PET counts determine the likelihood, while MR or CT similarities determine the kernel basis.

Our setting addresses the noisy and quality-varying cases rather than conventional independent-modality fusion because the learned representation is derived from the measured PET data. The measured sinogram and learned PET prior have different roles and error mechanisms, but they are statistically dependent. Within the proposed prior branch, grayscale and pseudocolour are two deterministic presentation views of the same preliminary PET image; Agent prompting and MedSAM2 decoding are successive processing stages rather than additional acquired modalities. We therefore describe the method as foundation-model prior fusion within kernel reconstruction and do not claim that it recovers information equivalent to an independently acquired MR or CT scan.

The dependence creates a specific failure path: low counts affect the MLEM image, which can alter the prompt objects and the MedSAM2 output, which then changes the KEM affinities. Prompt perturbations and prior ablations are therefore needed alongside ordinary reconstruction metrics. KEM and RBF affinities are established components [4, 3]; the contribution tested here is their use with a PET-adapted MedSAM2 prior, a dataset-aware prompt Agent, and a protocol that records prompting, decoder adaptation, and kernel reconstruction as separately auditable stages.

3. Method

3.1. Problem Formulation

SAM-KEM uses a low-count sinogram $\boldsymbol{y}$ as its measurement input and a continuous PET-like image $\boldsymbol{z}$ as its kernel feature source. The latter is derived from a preliminary reconstruction $\boldsymbol{x}_{\text{init}}$:

$$\boldsymbol{v}_g = C_g(\boldsymbol{x}_{\text{init}}), \quad \boldsymbol{v}_j = C_j(\boldsymbol{x}_{\text{init}}), \quad \mathcal{P} = f_{\text{Agent}}(\boldsymbol{v}_g, \boldsymbol{v}_j), \tag{2}$$

$$\boldsymbol{z} = \max_{o \in \mathcal{P}} f_{\text{SAM},\theta^*}(\boldsymbol{x}_{\text{init}}, o), \tag{3}$$

where C_g and C_j are fixed grayscale and jet display transforms, $\mathcal{P}$ is the validated set of per-object boxes and signed points, the maximum is pixelwise over continuous object logits, and θ^* denotes the adapted SAM parameters. The paired displays contain identical underlying PET samples and therefore constitute presentation views, not independent information sources. If the case is outside the frozen Agent route or Agent validation fails, $\mathcal{P}$ is supplied by the deterministic component rule. Because $\mathcal{P}$ conditions $\boldsymbol{z}$, both belong to one prior-generation branch. The final reconstruction is written as

$$\hat{\boldsymbol{x}} = \tilde{\boldsymbol{K}}(\boldsymbol{z})\boldsymbol{\alpha}^{(N_{\text{KEM}})}, \tag{4}$$

where the coefficients $\boldsymbol{\alpha}$ are updated under the Poisson likelihood of $\boldsymbol{y}$ and the sparse normalized kernel $\tilde{\boldsymbol{K}}(\boldsymbol{z})$ is fixed during KEM. The two inputs are dependent but have different functions: $\boldsymbol{y}$ determines data fidelity, while $\boldsymbol{z}$ determines kernel affinities and may be affected by localization or decoder errors. Figure 1 contrasts this PET-only prior-guided formulation with the principal alternative reconstruction strategies.

The architecture contains three explicit integration operations. First, the VLM jointly interprets the paired presentation views at the prompt-decision level. Second, the continuous logits from independently prompted objects are aggregated by a pixelwise maximum to form one prior image. Third, the prior determines the sparse kernel basis in which the measured-data likelihood is optimized. None of these operations is image-domain post-fusion of independently reconstructed endpoints.

For the dataset-aware Agent, f_{VLM} emits schema-validated per-object boxes and points after jointly inspecting the paired PET displays. The tool validates every coordinate before decoder execution, aggregates continuous object logits by pixelwise maximum, and uses a deterministic component prompt after invalid output or abstention. Neither path alters the KEM update or adds a post-reconstruction fusion term.

3.2. Low-Count Observation Model

PET projection data $\boldsymbol{y} = \{y_i\}$ are appropriately modeled as independent Poisson random variables, leading to the log-likelihood function [3]:

$$L(\boldsymbol{y} \mid \boldsymbol{x}) = \sum_{i=1}^{N_{\text{det}}} y_i \log \bar{y}_i - \bar{y}_i - \log y_i!, \tag{5}$$

where N_{det} is the number of measurement bins. The expectation of the projection data, $\bar{\boldsymbol{y}}$, relates to the unknown image $\boldsymbol{x}$ via:

$$\bar{\boldsymbol{y}} = \boldsymbol{P}\boldsymbol{x} + \boldsymbol{r}, \tag{6}$$

where $\boldsymbol{P}$ denotes the PET system matrix and $\boldsymbol{r}$ represents the expectation of random and scattered coincidence events. In low-count PET reconstruction, fewer detected true coincidences lead to a lower signal-to-noise ratio in the Poisson measurements.

For both datasets, low-count observations are generated from the nonnegative reference activity image rather than by adding a noise array to a reconstructed image or fixed-scale sinogram. Let $\boldsymbol{A}$ denote the forward operator and $\boldsymbol{p} = \boldsymbol{A}\boldsymbol{x}_{\text{ref}}$ the noise-free projection. For a requested expected true-event count N_{true}, we use

$$c = \frac{N_{\text{true}}}{\sum_i p_i}, \qquad \boldsymbol{\lambda} = c\boldsymbol{p} + \boldsymbol{r}, \qquad \boldsymbol{y} \sim \text{Poisson}(\boldsymbol{\lambda}), \tag{7}$$

where $\boldsymbol{r}$ is the declared background expectation. Reconstruction uses the same scaled operator $\boldsymbol{P} = c\boldsymbol{A}$ and background term $\boldsymbol{r}$ as the simulator. Each subject, count condition, and realization is sampled independently with a stored random seed. The expected-count variable is therefore the controlled data-quality axis: reducing it changes the Poisson observation statistics while leaving the source image, forward model, reconstruction definitions, and evaluation support fixed. In the executed clean protocol, attenuation, scatter, randoms, detector normalization, point-spread response, time of flight, and dead time are not modeled, so $\boldsymbol{r} = \boldsymbol{0}$.

The maximum likelihood (ML) estimate of $\boldsymbol{x}$ maximizes the Poisson log-likelihood:

$$\hat{\boldsymbol{x}}_{\text{ML}} = \arg\max_{\boldsymbol{x} \geq 0} L(\boldsymbol{y} \mid \boldsymbol{x}). \tag{8}$$

A standard approach to solving Eq. (8) is the Expectation-Maximization (EM) algorithm [1]. Ten MLEM iterations provide the preliminary image $\boldsymbol{x}_{\text{init}}$ used by the prompt and prior branch.

3.3. Dataset-Aware PET Prompt Agent

The formal Agent receives no MR, CT, target mask, reference image, or manual annotation. Its only image input is the ten-iteration MLEM image rendered as a deterministic side-by-side grayscale and jet-pseudocolour display. The two transformations change visual salience for the VLM but contain no additional measured information. Qwen3.7-Max is queried once per routed case with temperature zero and a strict JSON response contract.

The BrainWeb instruction describes a simulated head activity distribution and requests three or four salient uptake objects, each represented by a half-open box and one positive point, with no negative points. The HECKTOR instruction describes noisy head-and-neck PET and requests one lesion-centred object with one positive point and one to three background negative points outside the box. These dataset-specific differences were frozen before formal execution because the relevant uptake patterns and the role of background exclusion differ between the two cohorts. The primary routing policy uses the Agent for BrainWeb at counts up to 10^5 and DET at 2×10^5; HECKTOR uses the Agent throughout. The archived 10^5–10^6 experiment routed only BrainWeb 10^5 to the Agent, and the lower-count extensions retain the same prompt rules without prompt retuning.

The parser rejects duplicate keys, malformed object identities, noninteger or nonfinite coordinates, degenerate or out-of-bounds boxes, duplicate or overlapping signed points, and arm-specific cardinality violations. Each completed routed case has one logical prompt request; each low-count execution attempt permits at most one additional HTTP attempt for a transport error and caches exact-input receipts. Abstention or schema failure invokes the predeclared deterministic connected-component prompt and remains in the Agent arm under intention-to-treat analysis. The archived execution also retained transport fallbacks; the 20K extension instead pauses on persistent transport failure and preserves its transaction for recovery, separating service outages from completed policy decisions. After connectivity is restored, transport-failed transactions may be resumed without rerunning published valid or semantic-fallback decisions; cumulative HTTP accounting includes the preserved failed attempts as well as the completed transaction. Valid objects are decoded independently with the frozen dataset-specific MedSAM2 decoder, and their continuous logits are combined by pixelwise maximum before prior normalization and kernel construction. The Agent therefore performs prompt-level fusion of two presentation views and modifies only the prompt supplied to the learned prior branch; it neither selects among reconstruction outputs nor blends an Agent-derived image into the KEM endpoint.

3.4. Offline Adaptation and MedSAM2 Continuous Prior

The prior generator starts from MedSAM2 Hiera-T [21]. Its image, prompt and memory encoders remain frozen; adaptation updates only the native decoder path contributing to the PET-supervised single-mask output. The IoU/object-score heads and unused multimask hypernetwork branches are frozen, and no auxiliary reconstruction network is added. Reference PET is offline supervision, not an inference input or a manual segmentation label. One dataset-specific checkpoint is used across all four primary counts; checkpoints are not chosen separately for each dose or case.

The two datasets retain different, explicitly versioned adaptation histories. The HECKTOR primary checkpoint uses the historical L1 objective, trained jointly at 100K, 200K, 500K and 1M counts with equal-probability sampling. Its fixed budget is 8,000 single-image AdamW updates (20 epochs of 400

updates), learning rate 10^{-4} and weight decay 10^{-4}, with no best-epoch selection. Training boxes are the cached historical single-box proposals from anchor realization zero; other realizations at the same image and count share that box. The 20K/50K HECKTOR evaluations do not retune this decoder.

The revised BrainWeb checkpoint continues its historical PET-adapted initialization using deterministic multi-object PET prompts on the 16 training models. The first stage uses 24,576 updates balanced across models and six counts (20K–1M), with an L1 intensity term, a normalized finite-difference term and a local-affinity term:

$$\mathcal{L}_{\text{str}} = \|z - x_{\text{ref}}\|_1 + 0.1\mathcal{L}_{\Delta} + 0.05\mathcal{L}_{\text{aff}}. \tag{9}$$

Both structural terms use independently min–max-normalized prediction and target, with dynamic range floored at 10^{-6}. For offsets $(0,1),(1,0),(0,2),(2,0),(1,1),(1,-1)$, $\mathcal{L}_{\Delta}$ averages the absolute discrepancy of finite differences over valid pixel pairs, and $\mathcal{L}_{\text{aff}}$ averages the discrepancy of their affinities $\exp(-d^2/0.02)$. A second 24,576-update continuation uses the four primary counts and

$$\mathcal{L}_{\text{BW}} = \mathcal{L}_{\text{str}} + 0.3\,[1 - \text{SSIM}(z, x_{\text{ref}})]. \tag{10}$$

SSIM uses data range one and the full Gaussian-window map, including reflection boundaries. Both stages use batch one, AdamW with learning rate 10^{-5} and weight decay 0.01, and gradient clipping at one. The final fixed-step checkpoint is used at every primary count. A structural-only continuation shares the second stage's initialization, sample order and budget, separating the SSIM term from additional training. The available training pool comprises 2,815 PET planes; neither validation models nor target-derived prompts enter optimization.

The revised BrainWeb path fixes native output head zero at both training and inference and disables automatic stability-based multimask switching. Historical checkpoints can use that automatic fallback; results and component analyses from those checkpoints are not treated as matched ablations of the revised decoder. Box and point coordinates are transformed to the model's actual image grid, rather than to an assumed display resolution. At inference each validated object is decoded independently and the pixelwise maximum of the continuous object outputs forms z. It is not thresholded into a semantic mask and is never blended into the final reconstruction.

3.5. MedSAM2-Derived Kernel Construction

To incorporate the learned prior, we construct K from features ϕ derived from z:

$$\kappa(\phi_j, \phi_l) = \exp\left(-\frac{\|\phi_j - \phi_l\|^2}{2\sigma^2}\right), \quad l \in \mathcal{N}_J(j), \tag{11}$$

where ϕ_j is the feature at pixel j, σ is the kernel bandwidth, and $\mathcal{N}_J(j)$ is the spatial neighborhood of Manhattan radius J. The sparse kernel matrix is row-normalized:

$$\tilde{K} = \text{diag}(K\mathbf{1})^{-1}K. \tag{12}$$

The normalized kernel $\tilde{K}$ supplies a prior-dependent basis while the Poisson likelihood retains measurement consistency. Neighbouring pixels with similar prior values receive larger weights than pixels with dissimilar values.

The image is represented as $x = \tilde{K}\alpha$, and the kernel coefficients are estimated by maximizing:

$$\hat{\alpha} = \arg\max_{\alpha \geq 0} L(y \mid \tilde{K}\alpha). \tag{13}$$

The kernelized EM update rule [4] is:

$$\alpha^{(n+1)} = \frac{\alpha^{(n)}}{\tilde{K}^\top P^\top \mathbf{1}} \odot \tilde{K}^\top P^\top \left(\frac{y}{P\tilde{K}\alpha^{(n)} + r}\right), \tag{14}$$

where multiplication and division outside matrix products are element-wise and $\mathbf{1}$ is an all-ones projection vector. When $\tilde{K} = I$, this expression reduces to the standard MLEM update.

The stopping iteration N_{KEM} and kernel parameters follow the dataset-specific development and selection schedules in Section 4.2, then remain frozen during full validation. Historical NRMSE-selected parameters and the revised BrainWeb screen are distinct selection procedures; neither uses a per-case reference-dependent stopping rule at inference. The resulting kernel reconstruction is $x_{\text{KEM}} = \tilde{K}\alpha^{(N_{\text{KEM}})}$.

Figure 2 summarizes the complete executed architecture and its two coupled information paths: the measured sinogram supplies data fidelity, whereas the continuous MedSAM2 output supplies the fixed sparse reconstruction basis.

3.6. Kernel-Operator Interpretation and Sensitivity

The full RBF Gram matrix is symmetric and positive semidefinite. The implementation, however, applies spatial sparsification and row normalization; the resulting $\tilde{K}$ need not be symmetric, positive semidefinite, or an orthogonal projection. It is therefore most accurate to interpret $\tilde{K}$ as a prior-dependent linear smoothing operator inside the Poisson likelihood, rather than to claim that the normalized operator itself defines an RKHS.

For fixed neighborhood membership, an RBF weight changes smoothly with its feature difference. The scalar derivative has magnitude

$$\left|\frac{\partial}{\partial d}\exp\left(-\frac{d^2}{2\sigma^2}\right)\right| = \frac{|d|}{\sigma^2}\exp\left(-\frac{d^2}{2\sigma^2}\right) \leq \frac{e^{-1/2}}{\sigma}. \tag{15}$$

Thus small changes in the already normalized feature differences produce bounded changes in the unnormalized element-wise weights for fixed σ. This is not a bound with respect to the raw decoder output: per-prior min–max normalization can amplify perturbations when the raw dynamic range is small. Nor is it an end-to-end reconstruction bound, because row normalization couples the weights within each row, repeated KEM updates are nonlinear in the data, and the VLM–SAM mapping is nonlinear. Pipeline robustness must therefore be evaluated empirically rather than inferred from this local weight bound.

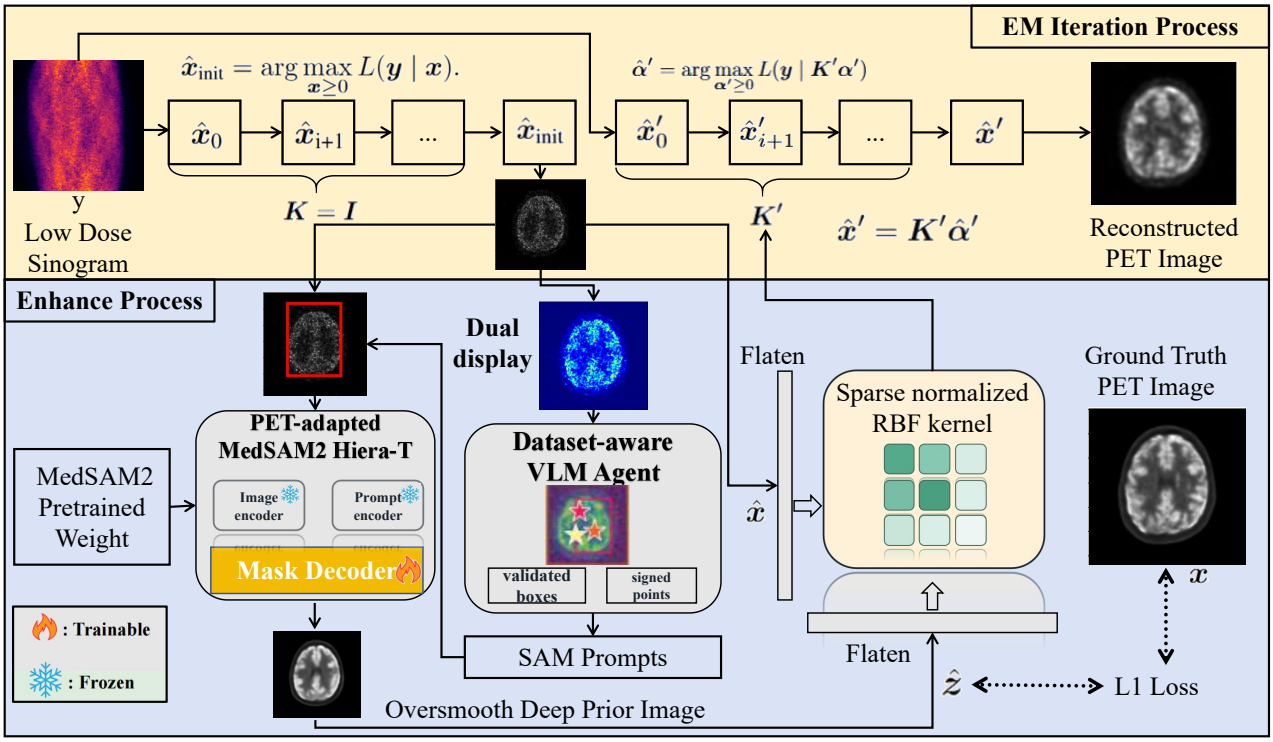


Figure 2: Agent–MedSAM2–KEM architecture for low-quality PET measurements, using the author-supplied schematic. Ten MLEM updates provide the paired PET displays for validated box/point prompting. Reference PET supervises the offline decoder: the illustrated L1 term is supplemented by the structural and SSIM terms for revised BrainWeb in Section 3.4. At inference, continuous object outputs form the normalized kernel feature; BrainWeb additionally smooths that feature before normalization. Coefficient-space KEM optimizes the measured-data Poisson likelihood. Invalid prompts use DET fallback. No MR/CT, tumour mask, auxiliary reconstruction head or output-level blending enters inference.

3.7. Implementation Details

Fixed dataset scale. Preliminary PET inputs and reference targets are divided by one frozen dataset-specific constant $C_{\mathcal{D}}$:

$$x_{\text{input}}^{\text{norm}} = x_{\text{input}}/C_{\mathcal{D}}, \qquad x_{\text{ref}}^{\text{norm}} = x_{\text{ref}}/C_{\mathcal{D}}. \tag{16}$$

BrainWeb and HECKTOR use different constants because their source-image intensity scales differ. The constants are training-only global maxima recorded in the executed configuration and frozen before validation reconstruction. They are development normalization constants in the respective source-image units, not scanner-calibrated physical activity constants. No pooled train-plus-validation min–max normalization is applied to the quantitative inputs or targets. For MedSAM2 encoder compatibility, the preliminary image is separately converted to a three-channel display image using per-image min–max scaling; this display conversion does not replace the fixed-scale L1 target.

Feature construction. The feature $\phi_j = \tilde{z}_j \in \mathbb{R}$ is the normalized continuous prior value at pixel j. Let $u = G_1 * z$ for revised BrainWeb and $u = z$ for HECKTOR, where G_1 is the one-pixel Gaussian feature filter. If $\max u - \min u > \epsilon$, use $\tilde{z} = (u - \min u)/(\max u - \min u)$; otherwise return an all-zero feature image. This per-prior min–max transform constructs the only path by which the decoder output enters reconstruction. The RBF kernel therefore groups pixels with similar predicted PET-prior intensity; it is not constructed from a four-dimensional feature vector or from a binary segmentation label. Spatial locality is enforced by the Manhattan-radius neighborhood $\mathcal{N}_J(j)$.

Kernel matrix construction and sparsity. For each pixel, weights are computed only within a Manhattan-radius neighborhood using a sliding window. At 128×128 resolution and $J = 2$, this neighborhood contains at most 13 pixels, giving approximately 2.13×10^5 nonzero entries rather than 2.68×10^8 dense entries. Construction scales as $O(HWJ^2)$. Runtime for this step is measured separately because it is a substantial component of the current implementation.

Formal Agent execution. The current route uses Qwen3.7-Max with temperature zero, a side-by-side grayscale–jet PNG, the dataset-specific instructions in Section 3.3, and one call per routed case. Raw responses are parsed without type coercion under the strict per-object schema; case transactions and exact-input receipt caching prevent repeated inference for verified completed cases. In the archived 10^5–10^6 run, the BrainWeb route is active only at 10^5 counts, whereas all four HECKTOR cells are routed to the Agent. The 20K/50K extensions additionally route both datasets at those counts, using the same dataset-specific instructions. Unrouted BrainWeb cells and all invalid, abstaining, or unavailable Agent responses use the deterministic component prompt. The requested provider alias, returned model metadata, prompts, display hashes, raw responses, validation outcomes, fallbacks and endpoint hashes

are retained in the formal audit package.

MedSAM2 configuration. The primary prior generator uses the MedSAM2 Hiera-T architecture. The image, prompt, and memory encoders and decoder heads disconnected from the selected single-mask PET output remain frozen during adaptation. The verified parameter inventory contains 38,962,498 parameters, of which 3,531,232 (9.1%) are trainable in both dataset-specific models. The pipeline contains no U-Net or additional reconstruction head. At inference, the frozen prompt encoder embeds the box and conditions the mask decoder, and the continuous output is retained instead of being thresholded or interpreted as a tumor segmentation.

The complete SAM-KEM reconstruction pipeline is summarized in Algorithm 1 and illustrated in Fig. 2.

Algorithm 1 Agent-Routed MedSAM2-Prior-Guided Kernel EM Reconstruction

Require: Low-count sinogram $\boldsymbol{y}$, scaled system matrix $\boldsymbol{P}$, background expectation $\boldsymbol{r}$
Require: Dataset-specific adapted MedSAM2 model $\text{MedSAM2}_{\theta^*}$
Require: Frozen Agent routing policy, strict response validator, and deterministic component fallback
Require: Hyperparameters: N_0, N_{KEM}, σ, J, feature filter G
Ensure: Reconstructed PET image $\hat{\boldsymbol{x}}$

Phase 1: MLEM Initialization
Initialize $\boldsymbol{x}^{(0)} \leftarrow \mathbf{1}$
for $n = 1$ to N_0 **do**
  $\boldsymbol{x}^{(n)} \leftarrow \text{MLEM_Update}(\boldsymbol{x}^{(n-1)}, \boldsymbol{y}, \boldsymbol{P}, \boldsymbol{r})$
end for
$\boldsymbol{x}_{\text{init}} \leftarrow \boldsymbol{x}^{(N_0)}$
Phase 2: Dataset-Aware Prompt Policy
if (dataset, count level) is Agent-routed **then**
  $\boldsymbol{I} \leftarrow C_{\text{gray+jet}}(\boldsymbol{x}_{\text{init}})$
  $\mathcal{R} \leftarrow \text{VLM-Agent}(\boldsymbol{I}, \text{dataset-specific instruction})$
  $\mathcal{P} \leftarrow \text{ValidateSchema}(\mathcal{R})$
  if $\mathcal{R}$ abstains, fails transport, or violates the schema **then**
    $\mathcal{P} \leftarrow \text{DET_COMPONENT}(\boldsymbol{x}_{\text{init}})$ and retain Agent failure
  end if
else
  $\mathcal{P} \leftarrow \text{DET_COMPONENT}(\boldsymbol{x}_{\text{init}})$
end if
Phase 3: MedSAM2 Prior Generation
for each object $o \in \mathcal{P}$ **do**
  $\boldsymbol{z}_o \leftarrow \text{MedSAM2}_{\theta^*}(\boldsymbol{x}_{\text{init}}, o)$
end for
$\boldsymbol{z} \leftarrow \text{PixelwiseMax}_{o \in \mathcal{P}}\, \boldsymbol{z}_o$
Phase 4: MedSAM2-Derived Kernel & KEM Reconstruction
$\tilde{\boldsymbol{z}} \leftarrow \text{PerPriorMinMax}(G * \boldsymbol{z})$ ▷ feature filter only
Set scalar features $\boldsymbol{\phi} \leftarrow \tilde{\boldsymbol{z}}$
for each pixel pair (j, l) with $l \in \mathcal{N}_J(j)$ **do**
  $\boldsymbol{K}_{jl} \leftarrow \exp\left(-\frac{\|\boldsymbol{\phi}_j - \boldsymbol{\phi}_l\|^2}{2\sigma^2}\right)$
end for
$\tilde{\boldsymbol{K}} \leftarrow \text{NormalizeRows}(\boldsymbol{K})$
Initialize $\boldsymbol{\alpha}^{(0)} \leftarrow \mathbf{1}$
for $n = 1$ to N_{KEM} **do**
  $\boldsymbol{\alpha}^{(n)} \leftarrow \text{KEM_Update}(\boldsymbol{\alpha}^{(n-1)}, \boldsymbol{y}, \boldsymbol{P}, \boldsymbol{r}, \tilde{\boldsymbol{K}})$
end for
$\boldsymbol{x}_{\text{KEM}} \leftarrow \tilde{\boldsymbol{K}}\boldsymbol{\alpha}^{(N_{\text{KEM}})}$
return $\hat{\boldsymbol{x}} \leftarrow \boldsymbol{x}_{\text{KEM}}$

4. Experiments

The evaluation separates four questions: comparison with iterative and learned methods, sensitivity to measurement counts, matched decoder/kernel changes, and the incremental prompt-policy effect. The grayscale/jet pair is not independently ablated; an Agent–DET contrast tests the complete policy, not the causal value of either display view.

4.1. Data, Splits and Simulation

We use subject-disjoint training and validation groups, but no untouched test cohort: available sources had already been accessed during earlier development. All results are therefore internal development-validation estimates.

BrainWeb. The 20 source models [25] are split into 16 training and four validation models, with 2,815 and 696 planes, respectively; all planes from one model remain in one group. Each complete 256×256 square is reduced to 128×128 by exact 2×2 mean pooling. The idealized pixel-driven parallel-beam operator has 180 views and 182 detector bins. Independent Poisson observations are stored at 20K, 50K, 100K and 200K expected true-event counts, with three realizations per plane. Thus each count has 2,088 case keys and the primary cohort has 8,352 keys, but only four independent validation models.

HECKTOR. HECKTOR 2021 [26] supplies 224 reconstructed head-and-neck FDG-PET/CT examinations, split into 179 training and 45 validation patients. The complete selected PET square is resized to 128×128 without a content crop; one plane per validation patient and three Poisson realizations give 135 keys per count and 540 across the primary counts. Tumour masks are not used for selection, prompting, training, metrics or case choice. The PET channel is the reference for the same simulated projection operator; this is clinical-image-derived pseudo-sinogram validation, not acquisition of low-dose clinical coincidence events. CT is used only by explicitly labelled anatomical comparators.

Both simulations omit attenuation, scatter, randoms, detector normalization, point-spread response, time of flight and dead time, with $\boldsymbol{r} = \mathbf{0}$. The counts are controlled measurement-quality proxies, not injected activities or calibrated effective doses. The 500K/1M observations remain historical high-count references in Appendix A. Training-only scale constants are $C_{\text{BW}} = 5.698180198669434$ and $C_{\text{HE}} = 34.93264389038086$; no validation-derived scale is fitted.

4.2. Compared Methods and Versioned Protocol

The nine comparators are MLEM, OSEM, EM-TV, Self-KEM, Bowsher-KEM, MR-KEM (BrainWeb) or CT-KEM (HECKTOR), and independent CNN-BPNet, FBP-Net and PET ADMM-Net adaptations. All methods use identical stored measurements, reference images, forward/adjoint operators and evaluation support for each case. SelfKEM uses min–max-scaled MLEM-10 features; the anatomical kernels use registered MR or CT, which the proposed method does not receive. Bowsher-KEM selects anatomy-similar neighbours within a Manhattan radius, row-normalizes the binary matrix and uses coefficient-space EM; it is not the classical penalized-likelihood Bowsher implementation. Classical 100K–1M settings were selected by subject-balanced validation NRMSE with SSIM tie-breaking and outward boundary shells; their 20K/50K extensions carry forward 100K settings without a new search.

The primary SAM-KEM rows use one revised BrainWeb decoder and one historical HECKTOR decoder, as specified in Section 3.4. Within each dataset–count cell, DET and the routed policy share the decoder, feature construction, kernel and stopping iteration and differ only in prompts. BrainWeb first smooths the continuous feature with a reflection-boundary Gaussian of standard deviation one pixel, then min–max normalizes it. Its frozen (σ, J, N) schedule is $(0.1, 6, 24)$, $(0.1, 4, 32)$, $(0.05, 4, 40)$ and $(0.05, 4, 60)$ at 20K, 50K, 100K and 200K. This feature smoothing occurs before kernel construction, not on the reconstructed endpoint. The schedule was selected in a 256-case development screen and local boundary extension, checked in a 384-case crossover, and then fixed for full validation without further tuning. The screen cases overlap the same four validation models; expansion to 8,352 keys is not independent confirmation. HECKTOR retains $(\sigma, J, N) = (0.1, 5, 20)$ without feature smoothing across all primary counts. The BrainWeb 200K route is DET, so its policy and DET endpoints coincide by construction.

The three learned baselines are explicitly independent implementations or static adaptations, not official reproductions. CNN-BPNet alternates five multiplicative back-projection updates and residual CNN blocks; FBP-Net uses a learned projection filter, fixed back-projection and residual denoiser; PET ADMM-Net unrolls ten data-consistency/proximal stages. At width eight their trainable parameter counts are 12,810, 6,083 and 25,620, respectively. One model per dataset is jointly count-conditioned over all six counts using batch 32, learning rate 3×10^{-5}, seed 20261013 and a 200-epoch ceiling. The minimum subject-balanced validation NRMSE checkpoint is retained; the preceding ten-epoch under-training screen is excluded. This is a common-observation and common-split comparison, not an equal-parameter or equal-training-compute comparison with the MedSAM2 decoder. An inapplicable MRI compressed-sensing ADMM upstream and license-restricted Neural KEM implementation are excluded rather than relabelled as PET baselines.

Iterative reconstruction uses sparse NumPy/SciPy CPU float64 operations, while learned inference and decoder adaptation use CUDA float32. All final image metrics are recomputed under a common contract. The support is the positive-sensitivity region of the complete-square operator, not a tumour or anatomical mask; here it covers the complete 128×128 square. NRMSE is the error norm divided by the reference norm, PSNR is $-10 \log_{10}(\text{MSE})$ with data range one, and SSIM is the mean full Gaussian-window map with $\sigma = 1.5$, population covariance and reflection boundaries. Metrics are first averaged within subject over planes and realizations, then equally across source models or patients. PSNR is averaged per case, not calculated from a pooled NRMSE.

4.3. Main Reconstruction Results

Table 1 groups all methods by dataset and uses the same three metrics in every count column. On BrainWeb, aligned MR-KEM has the lowest NRMSE and highest PSNR throughout; MR-KEM leads SSIM at 20K/50K and Bowsher-KEM at 100K/200K. The revised policy obtains 0.3503/27.19/0.8070 at 20K and 0.2590/29.86/0.8823 at 200K (NRMSE/PSNR/SSIM). At 100K and 200K it exceeds CNN-BPNet on all three metrics, but PET ADMM-Net still has higher SSIM at every count. Counting the policy and nine external comparators, but not DET as an additional external method, its NRMSE/PSNR/SSIM ranks are 4/3/5, 3/3/5, 3/3/4 and 3/3/4 from 20K to 200K. This does not establish replacement of aligned anatomy or dominance among PET-only methods on every metric.

On HECKTOR the retained routed policy leads the reported methods on all three metrics at each primary count. At 20K it obtains 0.2467/36.34/0.9177, compared with 0.2505/36.22/0.9151 for DET and 0.2616/35.82/0.9064 for PET ADMM-Net. The ranking is conditional on the completed comparator implementations and the shared selected-validation protocol.

4.4. Count-Quality Response

Figure 5 separates the count-quality response from the fixed-count method comparison. The cohort and observation model are unchanged along the count axis; decoder checkpoints stay fixed, while methods follow their declared count-specific or carry-forward solver schedules. Consequently the plot measures the performance of those schedules, not an isolated effect of counts with every stopping iteration held constant. The proposed method improves as counts increase, but the pattern of Agent benefit is dataset dependent rather than monotonic or universal. At BrainWeb 20K/50K, the revised DET prompt is better than the routed policy on all three metrics. At 100K the policy has a small NRMSE/PSNR advantage but lower SSIM; 200K is non-Agent DET routing. The historical HECKTOR policy remains favourable relative to its matched DET control at all four counts.

4.5. Matched Revised-Decoder and Mechanism Analyses

Table 2 reports the full-cohort BrainWeb structural continuation, SSIM-0.3 and prompt contrasts under identical fixed KEM schedules. The SSIM term increases DET SSIM at all counts relative to the equal-budget continuation, but slightly worsens

Table 1: Primary low-count reconstruction comparison. Entries are NRMSE↓/PSNR (dB)↑/SSIM↑, averaged within subject and then equally across subjects. Each method–count cell contains 2,088 BrainWeb cases (four source models) or 135 HECKTOR cases (45 patients). Bold denotes the best reported value. Model and parameter choices use the development-validation split; these comparisons are descriptive.

Dataset	Method	2×10^4	5×10^4	10^5	2×10^5
BrainWeb	MLEM [1]	0.4655/24.51/0.7283	0.3669/26.61/0.7711	0.3278/27.62/0.7958	0.3039/28.30/0.8372
	OSEM [2]	0.4784/24.27/0.7248	0.3714/26.49/0.7690	0.3298/27.56/0.7945	0.3063/28.23/0.8359
	EM-TV [3]	0.5462/23.50/0.7539	0.3625/26.98/0.8156	0.3034/28.50/0.8449	0.2774/29.33/0.8704
	Bowsher-KEM [5]	0.2717/29.27/0.8757	0.2189/31.16/0.9115	0.1982/32.05/**0.9270**	0.1776/33.00/**0.9409**
	SelfKEM [4]	0.5538/23.12/0.7655	0.3682/26.65/0.8083	0.3068/28.29/0.8317	0.2744/29.31/0.8637
	MR-KEM [4, 9]	**0.2533/29.95/0.8887**	**0.2124/31.52/0.9152**	**0.1975/32.20**/0.9262	**0.1768/33.16**/0.9397
	FBP-Net [12] (independent)	0.4520/24.80/0.6623	0.3847/26.20/0.7264	0.3476/27.09/0.7682	0.3176/27.88/0.8018
	CNN-BPNet [11] (independent)	0.3509/27.07/0.8107	0.3153/28.01/0.8391	0.2906/28.73/0.8589	0.2671/29.47/0.8777
	PET ADMM-Net [13] (independent)	0.3485/27.12/0.8213	0.3121/28.07/0.8478	0.2871/28.82/0.8669	0.2627/29.61/0.8852
	DET–SAM-KEM [4, 21]	0.3473/27.26/0.8083	0.3082/28.32/0.8389	0.2863/28.96/0.8632	0.2590/29.86/0.8823
	Agent/DET routed policy [4, 21, 22]	0.3503/27.19/0.8070	0.3100/28.27/0.8373	0.2860/28.98/0.8605	0.2590/29.86/0.8823
HECKTOR	MLEM [1]	0.3669/32.86/0.7799	0.2971/34.70/0.8562	0.2702/35.52/0.8878	0.2307/36.91/0.8941
	OSEM [2]	0.5969/28.64/0.5899	0.3845/32.46/0.7508	0.2912/34.88/0.8371	0.2341/36.78/0.8915
	EM-TV [3]	0.5305/29.71/0.7923	0.2702/35.55/0.9083	0.2063/37.88/0.9364	0.1807/39.04/0.9488
	Bowsher-KEM [5]	0.3133/34.25/0.8514	0.2411/36.53/0.9086	0.2124/37.64/0.9309	0.1804/39.11/0.9555
	SelfKEM [4]	0.2885/34.97/0.8986	0.2249/37.14/0.9309	0.2047/37.97/0.9419	0.1693/39.64/0.9592
	CT-KEM [4, 9]	0.2788/35.28/0.8988	0.2193/37.39/0.9369	0.1959/38.38/0.9504	0.1788/39.18/0.9562
	FBP-Net [12] (independent)	0.5225/29.81/0.5380	0.3715/32.76/0.6922	0.2961/34.73/0.7813	0.2463/36.33/0.8436
	CNN-BPNet [11] (independent)	0.2693/35.57/0.8923	0.2334/36.83/0.9147	0.2130/37.62/0.9267	0.1952/38.38/0.9371
	PET ADMM-Net [13] (independent)	0.2616/35.82/0.9064	0.2286/37.00/0.9245	0.2086/37.79/0.9324	0.1898/38.61/0.9406
	DET–SAM-KEM [4, 21]	0.2505/36.22/0.9151	0.1928/38.50/0.9452	0.1695/39.62/0.9557	0.1573/40.29/0.9613
	Agent/DET routed policy [4, 21, 22]	**0.2467/36.34/0.9177**	**0.1899/38.62/0.9465**	**0.1680/39.70/0.9563**	**0.1555/40.39/0.9619**

MR-/CT-KEM and Bowsher-KEM use additional aligned anatomy. Learned baselines are independent implementations trained jointly across six counts. BrainWeb SAM-KEM uses the revised head-0/SSIM-0.3 decoder and fixed count-specific KEM settings; HECKTOR retains the historical decoder and common KEM setting. Within each dataset–count, DET and policy differ only in prompts. BrainWeb 200K uses DET. Classical 20K/50K parameters carry forward the 100K selection. Historical decoder analyses and 500K/1M references appear in the appendices.

its NRMSE/PSNR at several counts. Thus it changes the noise–structure trade-off; its effect cannot be inferred by comparison only with the older, less-trained decoder. The revised single-output path also prevents stability-based switching to a branch without direct PET supervision. This corrects an output contract; it is not evidence that historical saved endpoints were numerically invalid.

Additional matched screens used 384 development cases, not new independent subjects. Direct object-stack and decoder-embedding features, prompt-local affinity, and background-negative completion did not surpass the scalar-maximum prior's matched SSIM. A 9,216-endpoint kernel screen compared scalar and 3×3 patch distances, full/9/17-neighbour support and common half/full stopping anchors for both prompts; no tested patch/kNN alternative improved matched scalar-control SSIM. A 1,920-endpoint prompt factorial compared original cached prompts, doubled box dimensions, an observed-MLEM peak positive point, and both changes. Box scaling increased Agent SSIM by 0.000298/0.001113/0.001594 at 20K/50K/100K, but remained below DET; at 100K it also worsened NRMSE and PSNR. Peak-point replacement contributed little. These are deterministic calibrations of cached prompts, not new VLM reasoning, and none is adopted in the primary pipeline.

The earlier eight-variant component replacements support adapted-prior and value-dependent-kernel effects for their historical single-box checkpoint. Their 71,136 endpoints, kernel-geometry diagnostics and fixed examples are retained in Appendix B rather than presented as matched ablations of the revised BrainWeb decoder. The present revised experiment does not repeat every historical component intervention and does not inherit its loss ratios or bootstrap evidence.

4.6. Matched Agent Effect and Statistical Scope

BrainWeb reuses all cached prompt decisions under the revised decoder; no new VLM calls were made for this revision. There are 6,264 Agent-routed keys, including 812 recorded fallbacks, and 2,088 DET-routed 200K keys. All remain in the intention-to-treat summaries; fallback and nonrouted cases account for 2,900 policy/DET prompt identities. Relative to DET, policy SSIM is lower by 0.001284, 0.001598 and 0.002673 at 20K, 50K and 100K, respectively. The first two counts also worsen NRMSE by 0.002918 and 0.001827; the 100K NRMSE change is −0.000325 with mixed model directions. Restricting to valid Agent responses does not remove the SSIM deficit, so transport or schema fallback is not a sufficient explanation.

New BrainWeb inference uses model-level deltas, exact enumeration of 16 sign flips and all 256 size-four model-bootstrap resamples. These are post-result exploratory analyses, with separate 12-test Holm families (four counts by three metrics) for each named comparison. All 84 reported decoder, prompt and

Table 2: Matched revised-BrainWeb full-validation analysis. Each cell contains NRMSE↓/PSNR (dB)↑/SSIM↑ for all 2,088 cases at that count. Continuation and SSIM-0.3 share their initialization, 24,576-update continuation budget, input order, head 0 and count-specific KEM setting. The last column is the policy-minus-DET SSIM difference under SSIM-0.3. These are selected-validation comparisons on the same four source models, not independent confirmation.

Count	Structural continuation, DET	SSIM-0.3, DET	SSIM-0.3, policy	ΔSSIM
20K	0.3447/27.32/0.8062	0.3473/27.26/0.8083	0.3503/27.19/0.8070	−0.001284
50K	0.3077/28.34/0.8370	0.3082/28.32/0.8389	0.3100/28.27/0.8373	−0.001598
100K	0.2853/29.00/0.8618	0.2863/28.96/0.8632	0.2860/28.98/0.8605	−0.002673
200K	0.2590/29.87/0.8809	0.2590/29.86/0.8823	0.2590/29.86/0.8823	+0.000000

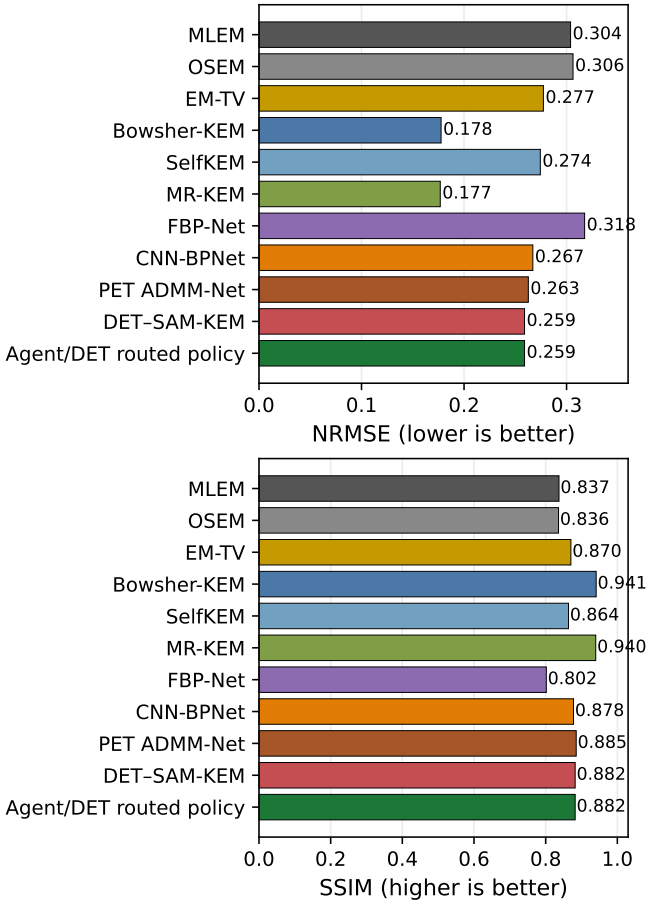


Figure 3: BrainWeb method comparison at 200K. Values match Table 1; lower NRMSE and higher SSIM are better. The learned baselines are independent adaptations. Policy equals DET at this count.

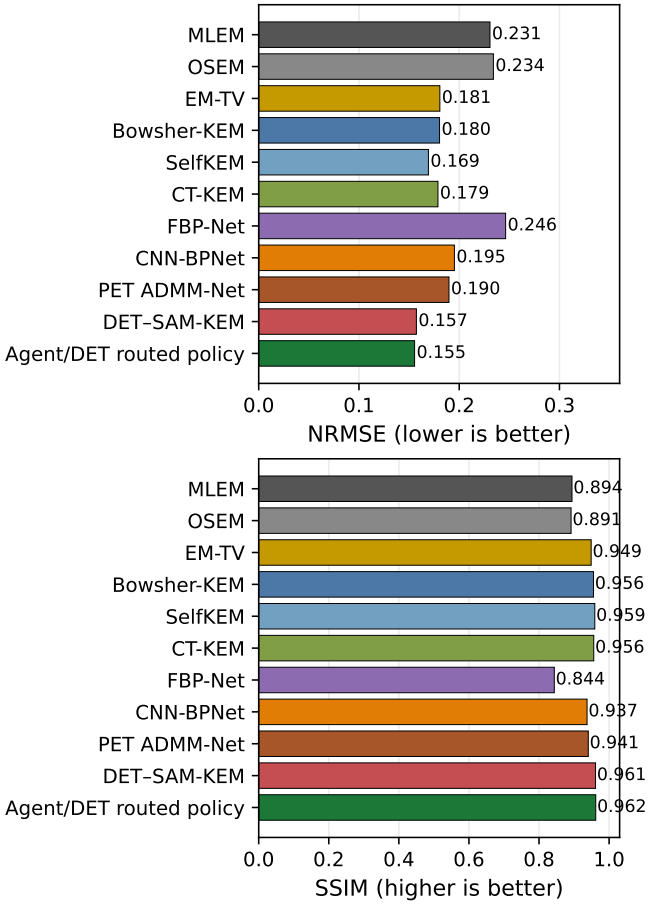


Figure 4: HECKTOR method comparison at 200K under the same metric conventions as Figure 3. The historical HECKTOR checkpoint is retained rather than selecting a different decoder for each count.

learned-baseline contrasts have Holm-adjusted $p = 1$; the minimum raw two-sided p is 0.125. Descriptive bootstrap intervals do not overcome the limited four-model sample or the reuse of validation data for development.

HECKTOR keeps its historical primary decoder, so the matched 20K extension remains directly applicable: Agent–DET changes are −0.003837 NRMSE, +0.1219 dB PSNR and +0.002639 SSIM. The 95% patient-bootstrap NRMSE interval is [−0.006249, −0.001909]; all three exploratory contrasts have $p_{\text{Holm}} = 6 \times 10^{-5}$ in the extension's original six-test family. Those six tests also include historical BrainWeb, which is not substituted for the revised analysis above. Historical 100K–1M Agent and prompt-component results are version-labelled in Appendix C. They identify background negatives as the clearest HECKTOR contribution, but cannot establish a prompt benefit independent of decoder version.

Operational audits remain relevant to the cached Agent policy: the original 2,628-call comparison used 201 deterministic fallbacks (7.65%). Among 16 fixed valid cases, only three had exact object agreement across a source response and two repeats; median box IoU was 0.063. Median API latency was 17.250 s versus 3.394 s for the archived rendering/decoder/kernel/KEM path combined. These are historical component timings, not a new end-to-end timing of the revised larger-radius BrainWeb

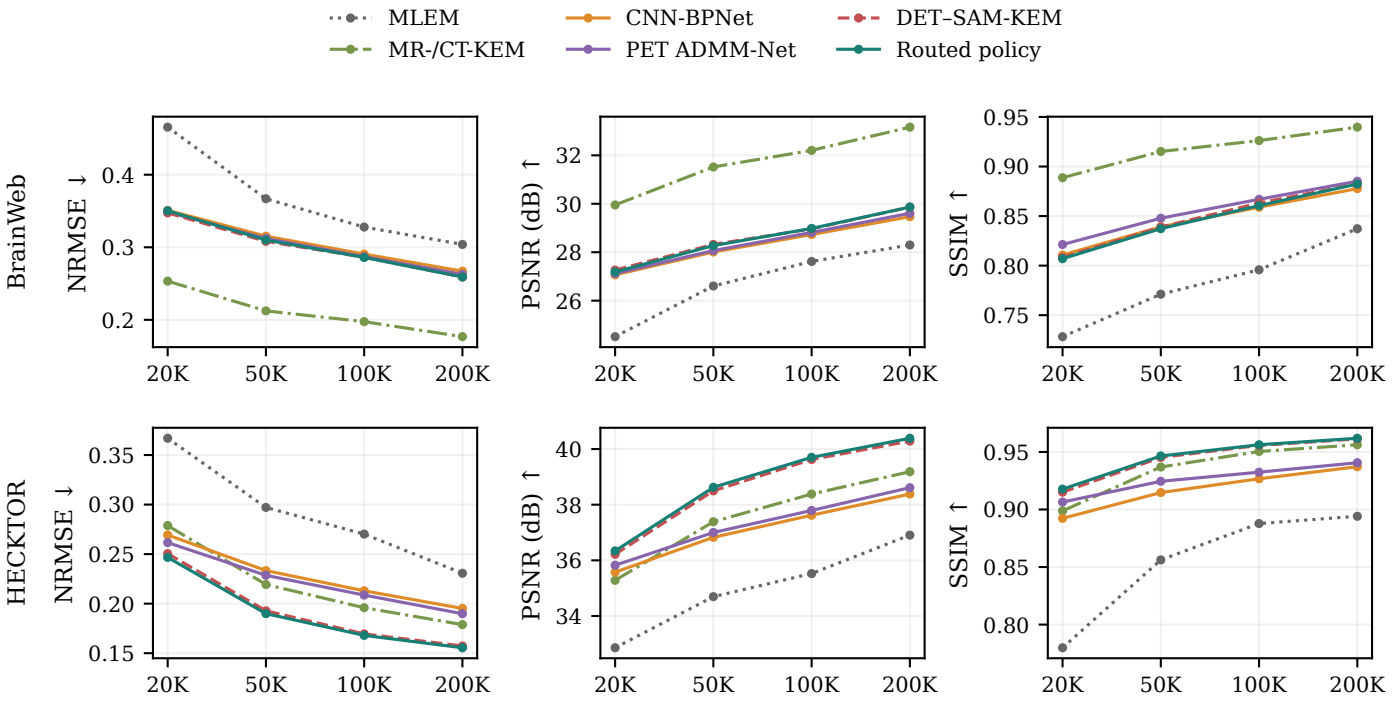


Figure 5: Primary count-quality response on the common validation cohort. Rows are BrainWeb and HECKTOR; columns are NRMSE, PSNR and SSIM. A representative subset is shown for legibility, with all methods and precise values in Table 1. Every curve uses that table's dataset-specific checkpoint and solver schedule. The display is not an independent experiment or an equal-training-compute comparison.

schedule. Valid JSON and temperature zero therefore do not imply spatial repeatability or negligible service cost.

4.7. Fixed-Case Visual Comparison

Figure 6 retains the four previously fixed case keys and all nine comparators; only the BrainWeb DET/policy panels are updated to the revised checkpoint and schedule. All displayed endpoints are archived KEM or selected comparator outputs, not selected anew by reconstruction quality. Every row uses one reference-defined grayscale window and labels PSNR/SSIM under the common metric contract. The Agent outline denotes method identity, not a winning endpoint.

5. Discussion

5.1. Reconstruction Core and Dataset-Dependent Prompting

The principal contribution is an explicit measurement–prior integration: a PET-adapted foundation-model output defines local kernel affinities, while coefficient-space EM retains the measured-data Poisson objective. The learned image is neither independent acquired anatomy nor an endpoint to blend with KEM. The primary comparison supports a competitive PET-only reconstruction path, especially on HECKTOR, but not dominance over aligned MR guidance or every learned PET comparator. At BrainWeb 100K/200K the revised model exceeds CNN-BPNet on all three metrics while remaining behind PET ADMM-Net in SSIM. These distinctions matter more than a single aggregate rank that hides metric trade-offs.

The Agent is not uniformly beneficial. On the retained HECKTOR model, dataset-aware lesion boxes and background negatives yield small favourable matched differences, including the exploratory 20K patient-level result. On the revised BrainWeb model, cached Agent prompts reduce SSIM relative to DET at every routed primary count, even after excluding schema fallbacks descriptively. The earlier concordant BrainWeb prompt effect therefore does not survive this decoder change. Prompt utility is an interaction with the learned prior and solver, not a property that can be transferred between checkpoints.

The follow-up geometry and kernel screens help localize the remaining issue without resolving it. Box enlargement recovers some SSIM, but not the DET result, while observed-PET peak points offer little improvement. Patch or nearest-neighbour sparsification is also not a general remedy under the tested settings. The current evidence does not identify a single causal explanation for the remaining BrainWeb gap; it excludes several simple remedies and discourages attributing it only to API failures. Historical component replacements support adaptation and value-dependent affinities for their own checkpoint, not every possible foundation-model prior.

5.2. Relevance to Low-Quality Information Fusion

Poisson count degradation changes the reliability of the preliminary image, its prompt objects and the learned kernel representation. The grayscale and jet renderings expose different visual presentations of those same noisy samples, not independent modalities. Object logits are aggregated inside the prior

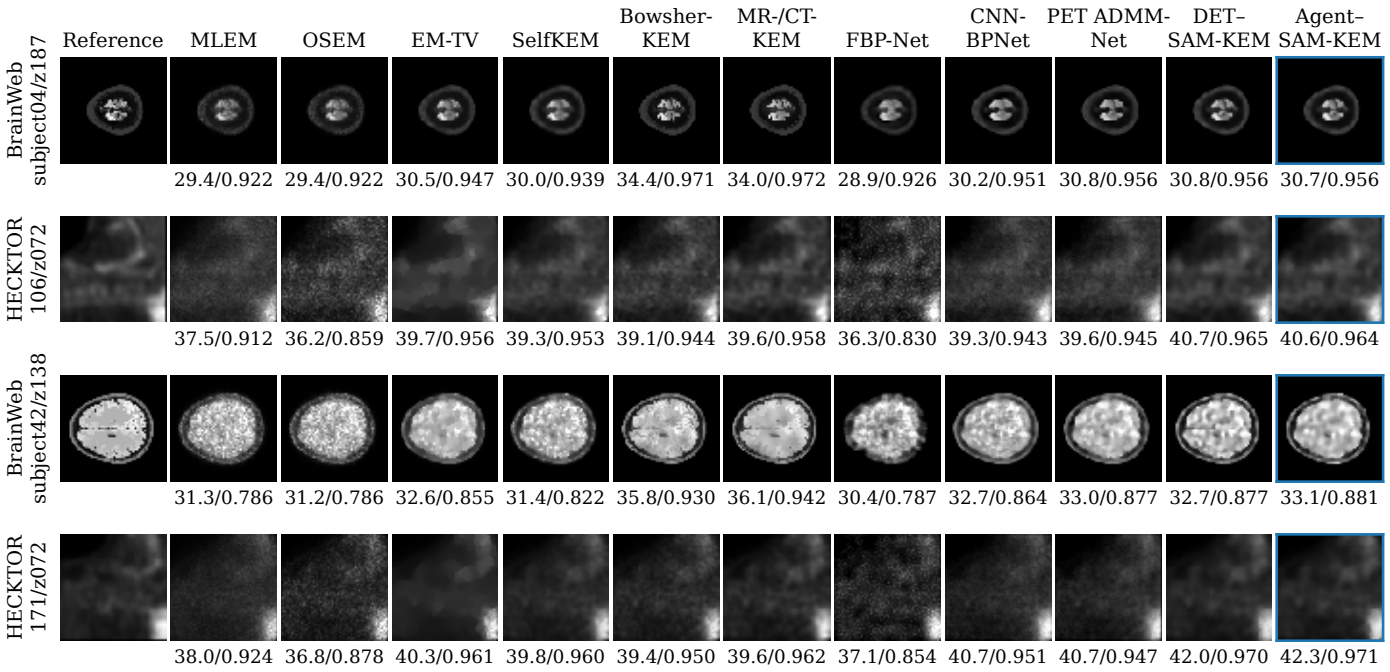


Figure 6: All-method comparison at 100K. Rows retain BrainWeb `subject04/z187` (realization 2), HECKTOR `106/z072` (realization 1), BrainWeb `subject42/z138` (realization 1), and HECKTOR `171/z072` (realization 2). Each row shares its measurements, reference and grayscale window (zero to reference maximum); labels are PSNR (dB)/SSIM. MR-/CT-KEM denotes the dataset-appropriate anatomical method. The three learned methods are independent adaptations. BrainWeb uses the revised decoder; HECKTOR retains its primary version. EM-TV arrays reproduce selected case metrics. Blue outlines identify the routed policy, not superiority.

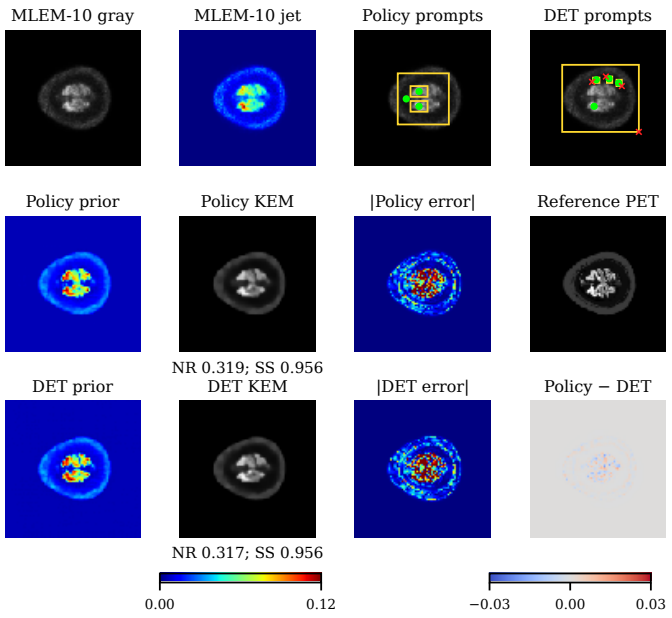


Figure 7: Revised BrainWeb prompt example, `subject04/z187`, 100K, realization 2. The same cached prompts now condition head 0 of the final SSIM-0.3 decoder. Reconstruction/reference panels share a grayscale range; the priors are independently scaled for display only. Errors share one scale, and the difference map is policy minus DET. This fixed example illustrates the current version, not a population-level benefit.

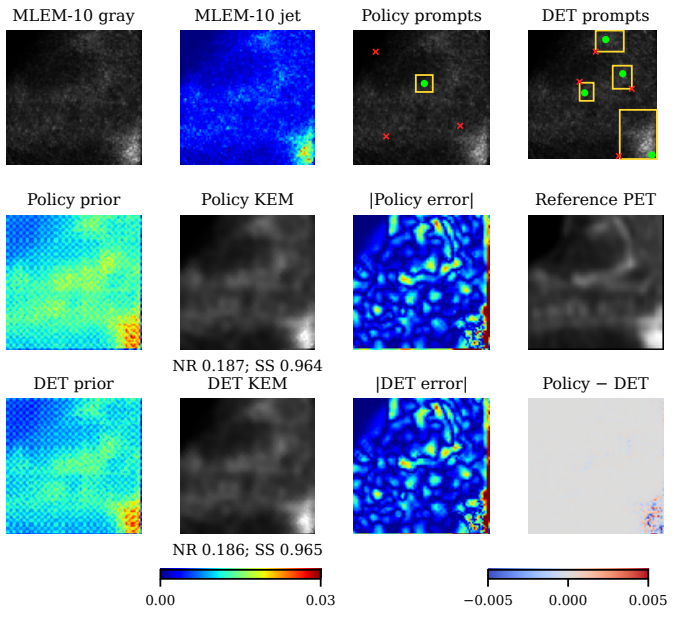


Figure 8: Retained HECKTOR primary-version prompt example, `106/z072`, 100K, realization 1. Panels show MLEM grayscale/jet, Agent/DET prompts, continuous priors, KEM endpoints, errors, reference and policy-minus-DET difference. Its historical decoder matches the HECKTOR primary table; it is not an example of the revised BrainWeb decoder.

branch, and that prior is integrated with measurement fidelity by changing the reconstruction basis. This is dependent-representation fusion for quality-varying measurements, rather than acquisition of missing MR/CT information. The retained likelihood makes the estimator explicit but cannot guarantee re-

covery when an erroneous kernel restricts the useful representation. Inspecting local affinities likewise does not by itself establish clinical explainability.

5.3. Limitations and Next Validation

Both cohorts use idealized two-dimensional projections; HECKTOR begins with clinical reconstructed images rather than raw coincidences. The simulations omit attenuation, scatter, randoms, detector response and three-dimensional geometry. Only four independent BrainWeb validation models are available, and repeated planes/noise realizations do not increase that independent sample size. Checkpoint, feature and stopping choices reuse development-validation outcomes; full-cohort execution after a screen is not an untouched external test. The revised decoder is a development selection, and no equal-training-compute superiority claim is made against the learned baselines.

Reference PET supervises decoder adaptation, so the experiment is not label-free pretraining or a demonstration that a VLM alone contains a PET reconstruction prior. Historical component interventions have not all been rerun under the revised BrainWeb decoder. The paired display has no grayscale-only or jet-only causal ablation, and the semantic repeatability, fallback and API-latency results constrain deployment claims. Hosted aliases do not pin immutable service revisions; archived receipts reproduce this study's prompt inputs and decisions, not future service behaviour. External cohorts, scanner-acquired measurements and version-matched component interventions are needed before generalization or clinical utility can be assessed.

6. Conclusion

SAM-KEM constructs a PET-derived continuous foundation prior, converts it into a sparse local KEM basis and estimates the final image against measured counts without MR/CT inference inputs or output-level blending. The revised BrainWeb and retained HECKTOR pipelines have complete 20K–200K comparisons under a unified three-metric contract. HECKTOR leads the tested comparators; revised BrainWeb improves over CNN-BPNet at 100K/200K but not over aligned MR guidance, and PET ADMM-Net retains higher SSIM. The Agent offers a dataset- and decoder-dependent refinement: its favourable HECKTOR effect does not extend to revised BrainWeb SSIM. Matched continuation and prompt controls make that distinction explicit rather than attributing every reconstruction improvement to VLM reasoning. The results support a reproducible foundation-prior/likelihood integration for low-quality PET measurements, with external and acquisition-realistic validation still required.

Data and Code Availability

BrainWeb models are available from `https://brainweb.bic.mni.mcgill.ca/brainweb/`; HECKTOR access remains subject to organizer terms at `https://hecktor.grand-challenge.org/Data/`. Third-party HECKTOR images are not redistributed with the manuscript. The provenance package records source splits, case keys, simulation seeds, raw prompt receipts, decoder hashes, solver plans and per-case metrics. The revised BrainWeb full-validation plan has SHA-256 prefix `3deb64d4163d`; its single SSIM-0.3 checkpoint has prefix `48acf5d8d3ad`. Both final-decoder and structural-continuation runs contain 8,352 cases and 16,704 DET/policy endpoints; final read-only review covers these and the two mechanism screens, totaling 44,544 overlapping endpoint instances rather than unique subjects. The versioned 88-cell result matrix, historical component evidence and paired-analysis test families remain separately indexed; they are not merged by selecting the best checkpoint at each count. Learned comparator code is labelled independent implementation/static adaptation, with restricted Neural KEM and the inapplicable MRI ADMM upstream excluded. The code repository is private at writing; a public archival DOI has not been assigned.

Appendix A. Archived Reconstruction and Higher-Count References

The archived comparison retains the 500K/1M reference conditions and the separately selected historical single-box SAM-KEM control. Its 100K/200K columns are retained only to identify that control and its earlier subject-level analysis; the primary low-count comparison is Table 1. The single-box control and the matched DET component-prompt pipeline have different prompts and kernel schedules and are not interchangeable.

Appendix B. Historical Component and Kernel Evidence

These archived experiments use the historical single-box decoder, automatic output-head fallback and separately selected 100K–1M kernel schedules. They do not use the revised BrainWeb head-0/SSIM-0.3 checkpoint. Their numerical records are retained unchanged, and their mechanistic conclusions are restricted to that historical implementation. The deterministic low-count prompt table likewise belongs to an earlier decoder sensitivity, not the current main matrix.

Appendix B.1. Historical Single-Box Prompt Baseline

Before development of the Agent/DET routed policy, the audited SAM-KEM baseline used Qwen3-VL-Plus to propose one outer box enclosing bright radiotracer uptake after $N_0 = 10$ MLEM iterations. This path generated prompts for historical decoder adaptation and the canonical endpoints of the archived matched component analysis; it is neither the revised BrainWeb training policy nor the final Agent inference policy. The baseline VLM receives the preliminary reconstruction after per-image min–max scaling, rounding to 8-bit intensity, and replication into three grayscale channels:

$$\boldsymbol{I}_{\mathrm{PET}} = C_{\mathrm{gray}}(\boldsymbol{x}_{\mathrm{init}}), \tag{B.1}$$

Table A.1: Reconstruction comparison at the four archived count levels on the common development-validation split. Every entry is NRMSE↓/PSNR (dB)↑/SSIM↑, averaged within subject and then equally across subjects. Methods are grouped by dataset. Hyperparameters and checkpoints were selected on the same validation outcomes; these are descriptive validation results, not independent-test estimates.

Dataset	Method	10^5	2×10^5	5×10^5	10^6
BrainWeb	MLEM [1]	0.3278/27.62/0.7958	0.3039/28.30/0.8372	0.2555/29.87/0.8720	0.2305/30.76/0.8929
	OSEM [2]	0.3298/27.56/0.7945	0.3063/28.23/0.8359	0.2562/29.84/0.8714	0.2372/30.57/0.8885
	EM-TV [3]	0.3034/28.50/0.8449	0.2774/29.33/0.8704	0.2375/30.68/0.8955	0.2098/31.76/0.9163
	Bowsher-KEM [5]	0.1982/32.05/**0.9270**	0.1776/33.00/**0.9409**	0.1594/33.98/0.9524	0.1470/34.69/0.9586
	SelfKEM [4]	0.3068/28.29/0.8317	0.2744/29.31/0.8637	0.2332/30.74/0.8954	0.2077/31.74/0.9143
	MR-KEM [4, 9]	**0.1975/32.20**/0.9262	**0.1768/33.16**/0.9397	**0.1558/34.27/0.9534**	**0.1413/35.10/0.9610**
	FBP-Net [12] (independent)	0.3476/27.09/0.7682	0.3176/27.88/0.8018	0.2878/28.74/0.8334	0.2722/29.23/0.8491
	CNN-BPNet [11] (independent)	0.2906/28.73/0.8589	0.2671/29.47/0.8777	0.2382/30.45/0.8998	0.2194/31.13/0.9139
	PET ADMM-Net [13] (independent)	0.2871/28.82/0.8669	0.2627/29.61/0.8852	0.2327/30.69/0.9075	0.2110/31.50/0.9222
	SAM-KEM [4, 21] (single box)	0.2976/28.46/0.8400	0.2709/29.29/0.8671	0.2389/30.40/0.8923	0.2161/31.27/0.9109
	Agent/DET routed policy [4, 21, 22]	0.2980/28.46/0.8410	0.2828/28.92/0.8522	0.2710/29.30/0.8614	0.2665/29.46/0.8652
HECKTOR	MLEM [1]	0.2702/35.52/0.8878	0.2307/36.91/0.8941	0.1908/38.57/0.9317	0.1725/39.44/0.9307
	OSEM [2]	0.2912/34.88/0.8371	0.2341/36.78/0.8915	0.1923/38.50/0.9307	0.1743/39.35/0.9295
	EM-TV [3]	0.2063/37.88/0.9364	0.1807/39.04/0.9488	0.1539/40.44/0.9579	0.1294/41.94/0.9659
	Bowsher-KEM [5]	0.2124/37.64/0.9309	0.1804/39.11/0.9555	0.1597/40.16/0.9584	0.1440/41.05/0.9628
	SelfKEM [4]	0.2047/37.97/0.9419	0.1693/39.64/0.9592	0.1355/41.56/0.9691	0.1143/43.04/0.9766
	CT-KEM [4, 9]	0.1959/38.38/0.9504	0.1788/39.18/0.9562	0.1562/40.39/0.9657	0.1408/41.28/0.9691
	FBP-Net [12] (independent)	0.2961/34.73/0.7813	0.2463/36.33/0.8436	0.2108/37.69/0.8898	0.1999/38.15/0.9054
	CNN-BPNet [11] (independent)	0.2130/37.62/0.9267	0.1952/38.38/0.9371	0.1770/39.23/0.9475	0.1709/39.54/0.9510
	PET ADMM-Net [13] (independent)	0.2086/37.79/0.9324	0.1898/38.61/0.9406	0.1662/39.78/0.9528	0.1547/40.40/0.9554
	SAM-KEM [4, 21] (single box)	0.1686/39.67/0.9563	**0.1522/40.57/0.9629**	**0.1219/42.54/0.9737**	**0.1051/43.86/0.9795**
	Agent/DET routed policy [4, 21, 22]	**0.1680/39.70/0.9563**	0.1555/40.39/0.9619	0.1464/40.93/0.9651	0.1429/41.14/0.9664

The higher-count 5×10^5 and 10^6 conditions are reference results, not primary low-count conditions. The $10^5/2\times10^5$ columns retain the separately selected historical single-box control for traceability; it is not the matched multi-object DET control in Table 1. Invalid Agent responses retain deterministic fallbacks. PSNR uses data range one and the same complete-square support. Decoder-version sensitivity is in Table C.5.

Table A.2: Subject-level consistency of the selected development-validation comparison. Each cell gives the subject-balanced mean ΔNRMSE (SAM-KEM minus comparator), followed by the number of subjects for which SAM-KEM has lower NRMSE. Negative values favour SAM-KEM. These are descriptive same-split comparisons because the reported validation outcomes were also used for hyperparameter selection.

Dataset	Comparator	10^5	2×10^5	5×10^5	10^6
BrainWeb ($n=4$)	MLEM	−0.0303; 4/4	−0.0331; 4/4	−0.0166; 4/4	−0.0144; 4/4
	SelfKEM	−0.0092; 4/4	−0.0036; 3/4	+0.0057; 0/4	+0.0084; 0/4
	MR-KEM	+0.1001; 0/4	+0.0940; 0/4	+0.0831; 0/4	+0.0748; 0/4
HECKTOR ($n=45$)	MLEM	−0.1016; 45/45	−0.0785; 45/45	−0.0688; 45/45	−0.0674; 45/45
	SelfKEM	−0.0361; 45/45	−0.0170; 41/45	−0.0136; 33/45	−0.0092; 33/45
	CT-KEM	−0.0273; 42/45	−0.0266; 43/45	−0.0343; 44/45	−0.0357; 43/45

where $C_{\text{gray}}(\cdot)$ creates a deterministic 128×128 RGB PNG whose three channels are identical. The VLM is prompted with a zero-shot instruction:

$$\boldsymbol{b} = \text{VLM}(\boldsymbol{I}_{\text{PET}}, \boldsymbol{p}_{\text{prompt}}), \tag{B.2}$$

where the executed UTF-8 Chinese task text, preserved verbatim in the archived run configuration, translates as "Enclose all bright clusters and return their smallest single outer bounding rectangle." The request implementation appends a strict JSON schema with a Boolean detection flag and integer $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$ coordinates on the 128×128 input, using inclusive left/top and exclusive right/bottom boundaries. The VLM returns $\boldsymbol{b}\in\mathbb{Z}^4$, which is parsed and validated against this convention. If no bright region is detected or the response cannot be parsed into a valid box, the pipeline applies a predeclared intensity-centroid algorithm to the same low-count preliminary reconstruction and constructs a 32×32 box around that centroid. Such a case remains recorded as a VLM failure even when fallback reconstruction succeeds. This is a zero-shot prompt: no example image or target box is supplied. The exact UTF-8 wording, provider model, temperature, timeout, request policy, centroid definition, and fallback-box dimensions are frozen in the run configuration and reproduced with their SHA-256 identities in the supplementary prompt-protocol file.

The matched component analysis in Appendix B compares this historical executed box with full-image, size-matched geometric-centre, and size-matched intensity-centroid prompts under otherwise frozen reconstruction settings.

Historical baseline prompt record. The Qwen3-VL-Plus single-box baseline receives the following prompt:

> *"Enclose all bright clusters and return their smallest single outer bounding rectangle."* (English transla-

tion)

The executed task string is Chinese and is retained byte-for-byte in the archived UTF-8 run configuration; the quotation above is its translation rather than a claim that an English prompt was executed. The request code appends the strict JSON schema and coordinate convention described in Appendix B.1. The baseline input is the 10-iteration MLEM reconstruction converted by deterministic per-image min–max scaling and rounding to a base64-encoded, three-channel grayscale PNG; no jet colour map or lossy JPEG compression is used on that historical path. The VLM is called through the DashScope OpenAI-compatible endpoint with model `qwen3-vl-plus`, temperature zero, a 120-s timeout, at most three attempts, and predeclared 0-, 2-, and 8-s pre-attempt delays. Returned coordinates are parsed as integer pixel coordinates and validated against the 128×128 half-open convention. A no-detection, unparsable, or geometrically invalid response triggers the predeclared intensity-centroid fallback on the same preliminary reconstruction, with relative threshold 0.5, a 32×32 fallback box, and the geometric centre used only when the thresholded intensity mass is zero; the event remains a VLM failure in the prompt record. The prompt is zero-shot: no example image or target box is supplied. The audited development cache contains 14,940 unique image–count keys: 14,044 from BrainWeb and 896 from HECKTOR across training and validation. The validation subset contains 2,964 keys (2,784 BrainWeb and 180 HECKTOR); one BrainWeb validation key used the recorded fallback and the other 2,963 used parsed Qwen boxes.

Appendix B.2. Frozen-Kernel Operating Regime

Whether the learned prior contributes useful local distinctions depends jointly on its normalized value differences and the selected RBF bandwidth. We therefore reconstructed the exact row-normalized kernel for every realization-zero validation prior and computed each row's inverse-participation count, $n_{\mathrm{eff}} = 1/\sum_l \tilde{K}_{jl}^2$. Table B.1 compares this count with the geometric degree d of the same Manhattan neighbourhood. The ratio is 0.9527–0.9947 on BrainWeb and 0.9521–0.9607 on HECKTOR; the three larger-count BrainWeb settings are especially close to a uniform local kernel. The mean normalized diagonal mass exceeds its uniform-neighbourhood counterpart by only 0.0039–0.0136 on BrainWeb and 0.0043–0.0048 on HECKTOR. This is a feature-level mechanism diagnostic, not evidence that the final KEM reconstructions are equal, because coefficient-space EM can amplify small matrix differences. It identifies a concrete failure hypothesis: at the selected bandwidths, much of the apparent benefit may arise from neighbourhood geometry rather than case-specific MedSAM2 feature values. The matched constant-feature endpoint below directly tests that hypothesis.

Appendix B.3. Ablation Experiments: Prior, Prompt, Decoder, and Kernel

We evaluated eight pure-KEM endpoints on the same 8,892 validation cases. All variants use the canonical observations, operators, support, frozen radius, bandwidth, stopping iteration, initialization, and metrics; only the declared prompt, decoder, or kernel feature changes. The canonical endpoint reproduces the frozen selected SAM-KEM result exactly for every case. The controls replace the Qwen prompt with a full-image, size-matched geometric-centre, or size-matched intensity-centroid box; replace the adapted decoder with released MedSAM2; replace the kernel feature with MLEM-10 or a deterministic cross-subject prior; or make all allowed local affinities uniform. Every variant has 8,892 finite endpoint rows and the final image remains the KEM estimate.

The completed low-count deterministic prompt extension is reported in Table B.3. On both datasets, adding a local box and signed points improves the subject-balanced NRMSE and SSIM relative to the full-image point control. The effect is larger on BrainWeb at 2×10^4 counts and remains positive at 5×10^4 counts; on HECKTOR, the local box is already beneficial and negative background points provide the best SSIM. These are prompt-mechanism results, not evidence for an API Agent effect, because all four arms use deterministic PET-derived prompts.

Table B.2 reports the predeclared primary NRMSE decision rule. The uniform replacement loses a median 0.387 of Q_ADAPT's observed improvement over MLEM and is worse in all eight dataset–count cells. The MLEM-feature and cross-subject controls are directionally compatible, with median losses of 0.257 and 0.136, respectively. These matched endpoints support a contribution from value-dependent, case-specific learned affinities even though the frozen kernels have effective-neighbour counts close to their uniform geometric support.

Replacing the adapted decoder with released MedSAM2 is worse in all eight cells and loses a median 0.314 of the observed improvement, supporting PET-specific decoder adaptation under the selected validation protocol. By contrast, the geometric-centre and intensity-centroid prompts have median losses of only 0.009 each; the centroid control matches or improves on Q_ADAPT in four of eight cells. The full-image replacement is also below the 0.10 localization threshold ($R = 0.048$). Thus the experiment does not establish Qwen localization as the source of the reconstruction gain. The cached Qwen box remains the prompt source for this historical matched baseline, while the formal prompt-policy experiment uses the Agent/DET route described above. Across both stages, the supported reconstruction core is the PET-adapted continuous prior and its value-dependent sparse KEM affinities.

HECKTOR patient-cluster bootstrap contrasts provide a consistency check rather than a confirmatory test. For the released-decoder, MLEM-feature, and uniform replacements, both NRMSE and SSIM intervals favour Q_ADAPT at all four count levels. For prompt-location controls, effects are small and mixed; the centroid replacement favours the control for NRMSE at all four HECKTOR counts. The deterministic seed (20260929) was chosen after the run for reproducible analysis and was not preregistered.

Figure B.1 shows the two 200k-count, realization-zero cases fixed before the matched analysis. The geometric-centre reconstruction is visually close to the Qwen-prompted endpoint,

Table B.1: Operating regime of the frozen SAM-KEM kernels on realization-zero validation priors. The effective-neighbour ratio is the rowwise inverse-participation count divided by the number of valid pixels in the same Manhattan neighbourhood, including the self-edge, averaged first within each prior and then across priors. A value near one indicates weights close to a uniform geometry-only local kernel; it does not by itself establish equal reconstruction endpoints.

	BrainWeb				HECKTOR			
Count	σ	Radius	n_{eff}	n_{eff}/d	σ	Radius	n_{eff}	n_{eff}/d
10^5	0.1	2	12.2295	0.9527	0.1	5	56.5754	0.9521
2×10^5	0.2	2	12.7165	0.9902	0.1	4	38.5643	0.9607
5×10^5	0.2	1	4.9398	0.9942	0.1	4	38.5273	0.9599
10^6	0.2	1	4.9422	0.9947	0.1	4	38.5233	0.9598

Table B.2: Matched component replacements under the frozen SAM-KEM settings. The primary statistic is the subject-balanced NRMSE improvement-loss ratio $R = (\text{NRMSE}_v - \text{NRMSE}_Q)/(\text{NRMSE}_{\text{MLEM}} - \text{NRMSE}_Q)$, evaluated in four count cells for each of two datasets. $R = 0$ retains the complete observed improvement over MLEM and $R = 1$ removes it. "Q wins" counts cells in which the canonical Qwen-prompted adapted prior has lower NRMSE. The predeclared materiality gates were $R \geq 0.20$ for the uniform replacement and $R \geq 0.10$ for prompt localization and decoder adaptation.

Question	Replacement	Variant	Q wins	Median R	Decision
Prompt extent	Full-image box	`FULL_ADAPT`	8/8	0.048	Small
Prompt location	Geometric-centre box	`CENTER_ADAPT`	7/8	0.009	Below gate
Prompt location	Intensity-centroid box	`CENTROID_ADAPT`	4/8	0.009	Below gate
Decoder adaptation	Released MedSAM2 decoder	`Q_RELEASED`	8/8	0.314	Supported
Feature content	MLEM-10 feature	`MLEM_FEATURE_MATCHED`	8/8	0.257	Compatible
Feature specificity	Cross-subject prior	`Q_SHUFFLED_MATCHED`	7/8	0.136	Compatible
Value-dependent affinity	Uniform local kernel	`UNIFORM_FEATURE_MATCHED`	8/8	0.387	Supported

Table B.3: Deterministic prompt ablation under the two most severely degraded low-count conditions on the validation split. Values are subject-balanced means of NRMSE↓/SSIM↑ over three realizations. The table is an extension experiment at 2×10^4 and 5×10^4 expected counts; it does not use VLM responses.

Dataset	Prompt	2×10^4	5×10^4	$\Delta\text{NRMSE}_{\text{box+pn−full}}$	$\Delta\text{SSIM}_{\text{box+pn−full}}$
BrainWeb	Full-image point	0.444522/0.774623	0.347244/0.812557	—	—
	Local box	0.440295/0.775288	0.346646/0.812641	—	—
	Local box + point	0.437199/0.776643	0.345029/0.813335	—	—
	Local box + pos./neg. points	**0.435515**/**0.777284**	**0.344758**/**0.813549**	−0.009007/ − 0.002486	+0.002661/ + 0.000992
HECKTOR	Full-image point	0.390516/0.785780	0.262642/0.882421	—	—
	Local box	**0.385303**/0.789750	0.259564/0.884736	—	—
	Local box + point	0.386052/0.789403	0.259801/0.884589	—	—
	Local box + pos./neg. points	0.385570/**0.790012**	**0.259563**/**0.884833**	−0.004946/ − 0.003079	+0.004232/ + 0.002412

The BrainWeb 5×10^4 NRMSE delta is −0.002486 and the SSIM delta is +0.000992; the corresponding 2×10^4 deltas are −0.009007 and +0.002661. Deltas are descriptive and are not VLM intention-to-treat effects.

whereas the released-decoder and uniform-kernel replacements produce more visible local differences in the HECKTOR example. These cases illustrate the interventions and are not used to infer population-level superiority; the decisions in Table B.2 use all 8,892 validation cases.

Appendix C. Historical Agent and Decoder Sensitivities

The following tables retain their original decoder and statistical families. Historical BrainWeb deltas and significance are not transferred to the revised primary model. HECKTOR primary-checkpoint contrasts remain relevant where the checkpoint, prompts and solver match. The archived formal analysis uses 10,000 subject-bootstrap samples, exact four-model or 100,000 Monte Carlo patient sign flips, seed 20261012, and its original ten-test Holm family. The 20K six-test extension and separate six-count-decoder 50K sensitivity are not pooled with that family.

Appendix C.1. Matched Agent Prompt Evaluation

The completed 20K extension contains 2,223 paired cases and 4,446 reconstruction endpoints under the historical dataset-specific decoders and unchanged prompt rules. On HECKTOR, the Agent–DET deltas are −0.003837 NRMSE (95% subject-bootstrap interval $[-0.006249, -0.001909]$), +0.1219 dB PSNR, and +0.002639 SSIM. All three contrasts have $p_{\text{Holm}} = 6.0 \times 10^{-5}$ in the extension's separate six-test family; 37/45, 38/45 and 42/45 patients favour the Agent for NRMSE, PSNR

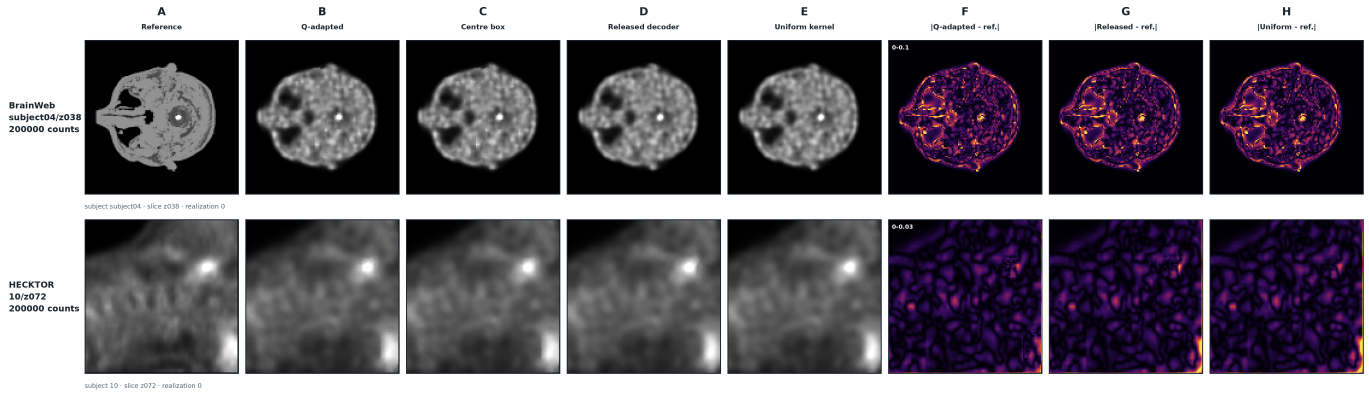


Figure B.1: Matched-control examples at 200k counts and realization zero, ordered from top to bottom as BrainWeb `subject04/z038` and HECKTOR `10/z072`. Columns A–E show the reference, canonical Qwen-prompted adapted-decoder KEM endpoint, size-matched geometric-centre prompt, released-decoder replacement, and uniform-local-kernel replacement. Columns F–H show absolute errors for the canonical, released-decoder, and uniform-kernel endpoints. Every reconstruction is a pure KEM output under the frozen settings. Grayscale windows are shared within each case (BrainWeb 0–0.30; HECKTOR 0–0.12); error windows are likewise shared within each case (0–0.10 and 0–0.03, respectively). Both rows were fixed before the matched analysis and are illustrative rather than inferential.

Table C.1: Completed 20K Agent extension relative to its matched DET control. Deltas are Agent minus DET; intervals resample subjects, not slices. These selected-validation contrasts form a separate exploratory six-test Holm family.

Dataset	Metric	Delta [95% CI]	p_{Holm}
BrainWeb	NRMSE	+0.00011 [−0.00058, +0.00086]	1.000
BrainWeb	PSNR (dB)	+0.002 [−0.013, +0.016]	1.000
BrainWeb	SSIM	+0.00011 [−0.00007, +0.00031]	1.000
HECKTOR	NRMSE	−0.00384 [−0.00625, −0.00191]	6.0×10^{-5}
HECKTOR	PSNR (dB)	+0.122 [+0.065, +0.195]	6.0×10^{-5}
HECKTOR	SSIM	+0.00264 [+0.00136, +0.00449]	6.0×10^{-5}

and SSIM, respectively. The BrainWeb 20K NRMSE delta is +0.000109 with mixed subject directions (2/4 favourable), while PSNR and SSIM changes are near zero; all three intervals cross zero and adjusted p-values are 1.0. Table C.1 reports these selected-validation contrasts without pooling them with the archived count family or the different-decoder 50K sensitivity.

The 20K intention-to-treat analysis retains all 477 semantic/schema fallbacks: 469/2,088 BrainWeb cases (22.46%) and 8/135 HECKTOR cases (5.93%). Its 2,223 logical requests comprise four reused smoke receipts and 2,263 new HTTP attempts associated with completed cases, including 44 within-transaction retries. Two stopped transport transactions add four preserved failed attempts, giving 2,267 new HTTP attempts for the full-run namespace; separate smoke and diagnostic requests are not included in that total. Transport recovery does not rerun published valid or semantic-fallback decisions, and the eight responses labelled `auto` by the service are retained rather than silently relabelled.

The archived 100K–1M formal comparison holds the observation, adapted decoder, kernel hyperparameters, and KEM endpoint fixed and changes only the prompt policy. It covers 8,892 validation cases and 11,520 endpoints, including the matched deterministic comparator for every routed Agent case. The Agent made 2,628 API calls: 2,427 responses passed the schema, while 201 cases (7.65%) used the declared deterministic fallback. All endpoints completed and all NRMSE and SSIM values were finite.

On BrainWeb at 10^5 counts, the subject-balanced Agent–DET delta was −0.001256 for NRMSE and +0.000466 for SSIM. All four independent subjects favoured the Agent for both metrics, but the exact sign-flip tests could not reach significance after Holm correction ($p_{\text{Holm}} = 0.25$) with only four subjects. On HECKTOR, all four count levels favoured the Agent for both metrics. Mean NRMSE reductions ranged from 0.000963 to 0.001779, and mean SSIM increases ranged from 0.000393 to 0.000583. All eight HECKTOR contrasts remained significant after Holm correction ($p_{\text{Holm}} \leq 10^{-4}$), with favourable-subject fractions of 0.756–0.867 for NRMSE and 0.889–0.956 for SSIM.

Prompt-component ablation localized the most consistent HECKTOR contribution to background negative points: adding them to the lesion box and positive point improved both metrics in all four count cells, with all eight Holm-adjusted tests significant. By contrast, adding the positive point to the box alone was adverse in several medium- and high-count cells. On BrainWeb, the box-only contrast retained almost the complete Agent effect, whereas point addition was near zero. These findings support a dataset-specific prompt mechanism rather than a universal claim that more prompt tokens are beneficial.

API and grounding audits expose the remaining reliability cost. Among 16 frozen cases with valid source responses, only three had exact object agreement across the source call and two repeats; median box intersection-over-union was 0.063. A later quota-restored run completed 100/100 calls over 20 frozen cases, showing that the execution path can be operationally stable, but the spatial repeatability result remained the relevant semantic limitation. Median API time was 17.250 s per Agent endpoint compared with 3.394 s for rendering, decoding, kernel construction, and KEM combined. Tables C.2 and C.3 report the inferential and audit evidence.

The prespecified 5×10^4 carry-forward sensitivity run with

Table C.2: Archived 100K–1M intention-to-treat effect of the dataset-aware prompt Agent relative to the deterministic component prompt. Deltas are Agent minus DET after averaging repeated realizations within subject. Negative NRMSE and positive SSIM favour the Agent. Confidence intervals use subject-level bootstrap resampling; p_{Holm} is the Holm-adjusted sign-flip p-value across the ten prespecified tests. The completed 20K extension has a separate test family (Table C.1).

Dataset	Counts	Metric	Subjects	Mean delta	95% CI	p_{Holm}
BrainWeb	10^5	NRMSE ↓	4	−0.001256	[−0.001986, −0.000526]	0.250
BrainWeb	10^5	SSIM ↑	4	+0.000466	[+0.000054, +0.001266]	0.250
HECKTOR	10^5	NRMSE ↓	45	−0.001494	[−0.001961, −0.001036]	1.0×10^{-4}
HECKTOR	10^5	SSIM ↑	45	+0.000583	[+0.000400, +0.000787]	1.0×10^{-4}
HECKTOR	2×10^5	NRMSE ↓	45	−0.001779	[−0.002467, −0.001146]	1.0×10^{-4}
HECKTOR	2×10^5	SSIM ↑	45	+0.000572	[+0.000374, +0.000799]	1.0×10^{-4}
HECKTOR	5×10^5	NRMSE ↓	45	−0.001311	[−0.001760, −0.000840]	1.0×10^{-4}
HECKTOR	5×10^5	SSIM ↑	45	+0.000446	[+0.000328, +0.000576]	1.0×10^{-4}
HECKTOR	10^6	NRMSE ↓	45	−0.000963	[−0.001370, −0.000550]	1.0×10^{-4}
HECKTOR	10^6	SSIM ↑	45	+0.000393	[+0.000300, +0.000491]	1.0×10^{-4}

Table C.3: Archived formal Agent execution, mechanism, reliability, and efficiency evidence, with the separate six-count-decoder 50K sensitivity. API time dominates the archived incremental latency. Prompt contrasts are computed only for valid Agent responses, whereas Table C.2 retains deterministic fallbacks under intention-to-treat analysis.

Evidence	Result	Interpretation
Formal execution	8,892 cases; 11,520 endpoints; 2,628 API calls	2,427 valid responses and 201 deterministic fallbacks (7.65%); no failed endpoint or non-finite metric
HECKTOR negative-point contrast	favourable NRMSE and SSIM in all four count cells; 8/8 Holm-adjusted tests significant	Negative background points are the clearest prompt-level mechanism of the HECKTOR gain
HECKTOR positive-point contrast	adverse at several medium/high-count cells	A positive point is not independently beneficial and must be interpreted within the complete prompt
Fixed-call repeatability	exact three-call agreement 3/16; median box IoU 0.063	Valid structured output does not imply stable spatial grounding
Stable supplemental API run	100/100 valid calls over 20 frozen cases	The execution path is feasible after quota recovery, but this does not remove semantic variability
5×10^4 carry-forward sensitivity	2,223 cases; 1,944 valid responses; 279 deterministic fallbacks (12.55%)	BrainWeb remains directionally favourable; HECKTOR effects attenuate and cross zero at this more severe count level
Median component time per Agent endpoint	render 0.153 s; API 17.250 s; decoder 0.276 s; kernel 2.638 s; KEM 0.432 s	Median end-to-end component sum 20.767 s; API inference is the main overhead

the separate six-count decoders additionally completed 2,223 paired cases and 4,446 endpoints, with 1,944 valid Agent responses and 279 deterministic fallbacks (12.55%). On BrainWeb, all four subjects again favoured the Agent for both metrics (ΔNRMSE = −0.001245; ΔSSIM = +0.000407), although the exact tests remained underpowered after correction. On HECKTOR, the mean directions remained favourable but attenuated (ΔNRMSE = −0.000362; ΔSSIM = +0.000146), and both confidence intervals crossed zero. These inferences apply to those six-count decoder checkpoints, not to the historical-checkpoint 50K replays used in the historical main table.

Decoder-update sensitivity.. We additionally replayed the cached BrainWeb 5×10^4 prompt decisions using a decoder trained with deterministic boxes on 30 slices from each of the 16 training models, jointly across six count levels. This selected development candidate uses the same 20-epoch, 402-update-per-epoch budget as its matched decoder-scale controls. Table C.5 holds the observations, cached Agent decisions, multi-object deterministic comparator, and KEM parameters fixed across decoder versions.

The updated Agent endpoint improves over the original Agent by 0.005932 NRMSE, 0.1312 dB PSNR, and 0.004376 SSIM. Relative to its updated deterministic comparator, the Agent retains improvements of 0.001416 NRMSE, 0.0454 dB PSNR, and 0.000388 SSIM. All four models favour the Agent on all three metrics, but the exact two-sided subject-level sign-flip tests yield $p = 0.125$ before multiplicity correction. The 100K replay changes this interpretation: BW30 gives NRMSE 0.30934, PSNR 28.0651 dB, and SSIM 0.83530, compared with 0.29797, 28.4618 dB, and 0.84097 for the historical four-count Agent checkpoint. The matched 50K replay of the historical four-count decoder gives NRMSE 0.32618, PSNR 27.6380 dB, and SSIM 0.82374, improving on BW30's 0.34037, 27.2055 dB, and 0.81697. At 200K, the deterministic route with the historical decoder gives 0.28278 NRMSE, 28.9216 dB PSNR, and 0.85217 SSIM, compared with BW30's 0.29328, 28.5430 dB, and 0.84742. Thus B-

Table C.4: Prespecified 5×10^4-count carry-forward sensitivity analysis. Deltas are Agent minus DET under intention-to-treat analysis; negative NRMSE and positive SSIM favour the Agent. The four tests form a separate Holm family from Table C.2.

Dataset	Metric	Subjects	Mean delta	95% CI	Favourable subjects	p_{Holm}
BrainWeb	NRMSE ↓	4	−0.001245	[−0.001984, −0.000808]	4/4	0.500
BrainWeb	SSIM ↑	4	+0.000407	[+0.000107, +0.000888]	4/4	0.500
HECKTOR	NRMSE ↓	45	−0.000362	[−0.000880, +0.000217]	30/45	0.500
HECKTOR	SSIM ↑	45	+0.000146	[−0.000331, +0.000474]	36/45	0.690

Table C.5: Decoder-version sensitivity with cached prompts and identical observations and KEM settings within each dataset–count cell. Entries are subject-balanced NRMSE↓, PSNR (dB)↑, and SSIM↑ from 2,088 BrainWeb cases (four models) or 135 HECKTOR cases (45 patients). BrainWeb 200K routes deterministically. Training coverage, count sampling and training prompts change jointly; these are exploratory version comparisons, not isolated sample-size effects.

Dataset / counts	Decoder	Prompt	NRMSE	PSNR	SSIM
BrainWeb / 50K	Original six-count	DET	0.34755	27.0429	0.81219
	Original six-count	Agent	0.34630	27.0743	0.81259
	BW30	DET	0.34179	27.1601	0.81658
	BW30	Agent	0.34037	27.2055	0.81697
BrainWeb / 50K	Historical four-count	DET	0.32685	27.6162	0.82345
	Historical four-count	Agent	0.32618	27.6380	0.82374
BrainWeb / 100K	Historical four-count	DET	0.29923	28.4125	0.84050
	Historical four-count	Agent	0.29797	28.4618	0.84097
	BW30	DET	0.30983	28.0425	0.83535
	BW30	Agent	0.30934	28.0651	0.83530
BrainWeb / 200K	Historical four-count	DET	0.28278	28.9216	0.85217
	BW30	DET	0.29328	28.5430	0.84742
HECKTOR / 50K	Low-count six-count	DET	0.18918	38.6523	0.94670
	Low-count six-count	Agent	0.18882	38.6697	0.94684
	Historical four-count	DET	0.19277	38.4971	0.94524
	Historical four-count	Agent	0.18993	38.6241	0.94650
HECKTOR / 100K	Historical four-count	DET	0.16947	39.6242	0.95568
	Historical four-count	Agent	0.16798	39.7003	0.95626
	Low-count six-count	DET	0.16733	39.7306	0.95638
	Low-count six-count	Agent	0.16716	39.7420	0.95656
HECKTOR / 200K	Historical four-count	DET	0.15726	40.2855	0.96132
	Historical four-count	Agent	0.15548	40.3867	0.96189
	Low-count six-count	DET	0.15481	40.4172	0.96162
	Low-count six-count	Agent	0.15466	40.4275	0.96191

W30 improves on the six-count low-count checkpoint but remains worse than the historical BrainWeb decoder on all three metrics at 50K, 100K, and 200K. At that development stage the historical BrainWeb checkpoint was retained; these BW30 values are archived sensitivities, not the revised BrainWeb primary results. Checkpoint provenance shows that the historical BrainWeb run sampled 2,695 distinct training planes over four count levels, whereas BW30 sampled 480 planes over six counts and used deterministic rather than cached VLM training boxes. These simultaneous changes prevent attributing the difference to training-set size alone. On HECKTOR, the matched historical-decoder 50K replay gives Agent NRMSE/PSNR/SSIM of 0.18993/38.6241/0.94650, compared with 0.18882/38.6697/0.94684 for the six-count decoder. The six-count decoder also gives Agent NRMSE/SSIM of 0.16716/0.95656 at 100K and 0.15466/0.96191 at 200K, improving absolute reconstruction quality over the historical checkpoint at all three tested counts. The matched Agent-minus-DET NRMSE differences are only −0.000169 and −0.000141, with subject-bootstrap intervals crossing zero; SSIM differences are +0.000181 and +0.000282. The decoder revision therefore changes both reconstruction quality and the magnitude of the prompt-policy effect; the historical Agent effect sizes cannot be transferred to a new checkpoint. This exploratory decoder-update sensitivity does not replace the current primary comparison or inherit the component-analysis claims.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI Codex (accessed in 2026) to assist with language editing, LaTeX organization, source-audit review, and deterministic figure code. After using this tool, the authors reviewed and edited the content, independently checked the cited sources and numerical

evidence, verified the diagrams against the executed method, and take full responsibility for the content of the publication.

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